Short-horizon mean reversion in cryptocurrency markets: a matched cross-market measurement

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August 2026

Abstract

We score 183 cryptocurrency pairs against 187 US stocks and ETFs at 15 minutes under one matched, strictly out-of-sample walk-forward. Directional mean reversion concentrates in crypto: the class-mean gap in out-of-sample AUC is +0.031 (95% CI [+0.027, +0.035]) as fixed ex ante, +0.022 excluding flat bars, and +0.011 [+0.008, +0.014] with both artifact adjustments applied at once — each clear of zero, reproduced by parameter-free sign statistics, and 4.4 standard deviations beyond an exact permutation null that preserves the full dependence structure. A frozen six-month holdout postdating every modelling choice retains the gap at +0.020; four fifths of the signal survives removing the crypto market factor; and a joint bootstrap caps any common stock-class effect at a fortieth of the crypto excess. US-listed funds whose net asset value is a crypto or metal price carry their underlying's reading, nulls included, while correlated instruments without the NAV link carry nothing — descriptive evidence that the signature travels with the price process rather than the listing venue. On the originating tape, reversal concentrates after aggressive-flow-driven moves, yet the gross edge peaks near 1.3 bp per trade against a 5 bp cheapest round-trip cost: compensated liquidity provision too small to be worth arbitraging away.

Keywords: market efficiency; mean reversion; price discovery; exchange-traded funds; order flow; limits to arbitrage

JEL classification: G14; G12; C58

1 Introduction

Market efficiency is a matter of degree [18, 22, 37]: different markets absorb information at different speeds, and the residual predictability left behind is a measurable signature of their microstructure. This paper measures that residual at the shortest horizons, at population scale, across today's two most different market structures. Scored under one matched, strictly out-of-sample walk-forward, 183 cryptocurrency pairs carry a pervasive 15-minute mean-reverting sign predictability that 187 US stocks and ETFs do not: small (2–3 points of AUC), decaying within hours, stable across five years, and surviving every artifact channel we can test, a six-month forward holdout, an exact permutation null, and the removal of the common crypto market factor (Section 3).

Two supporting analyses bound what the fact means. First, US-listed funds whose net asset value *is* a crypto or metal price reproduce their underlying's session-matched reading — nulls included — while instruments merely correlated with the same underlyings carry nothing; transmission is present from a fund's first months, through a futures-based NAV, at 2× leverage, and inside the sampling minute (Section 4). This is descriptive evidence, not identification: at one-minute correlation 0.982 a wrapper's bars nearly *are* its underlying's, so we read the panel as showing that efficiency statistics computed on a wrapper's tape describe the wrapped process, not the listing venue. Second, on the originating tape the reversal concentrates after moves that aggressive taker flow pushed and grows with flow intensity, yet bar-level flow adds nothing out of sample beyond the sign path — the signature, not the identification, of liquidity provision compensating overreaction. The gross edge peaks near 1.3 bp per trade against a 5 bp cheapest round-trip cost, so nothing here is capturable at spot costs: an effect big enough to measure and too small to be worth removing [22, 48] (Section 5).

The evidential hierarchy is stated once and held to. The population fact rests on a test fixed in design before the wide-universe run and is the firmest claim; the wrapper panel is an observational pattern on eight underlying families, strong in direction and without formal identification; the mechanism evidence constrains accounts without selecting one. The measurement instrument is deliberately minimal — a three-parameter constrained distributed-lag logit used as a low-variance device, not a model of markets. Nothing rests on it: an unconstrained logit and parameter-free sign statistics reproduce the contrast, and its derivation and baselines are confined to Appendix A.

Relation to the literature The paper sits in the microstructure literature on short-horizon reversal and liquidity provision. In equities, reversal at daily and intraday horizons has long been read as compensation for absorbing order-flow imbalances [21, 29, 36, 42]: strongest after high-volume moves [11], forecastable from signed order imbalance [12], tied to illiquidity in the cross-section [2], estimated structurally as inventory-driven price pressure [27], and competed down over time by professional intermediaries [13, 14]. We measure where that competition has and has not done its work across today's market structures. The ETF literature has used the creation–redemption link mainly in the opposite direction from our panel: tracing how arbitrage transmits non-fundamental shocks *into* underlying securities [4], characterising deviations of wrapper prices from net asset value [40, 44], and asking how ETF activity affects the underlying's efficiency [20] or where price discovery sits among linked instruments [26]; we take the NAV constraint as given and ask, descriptively, whether a wrapper carries its underlying's *conditional* short-horizon statistic. On the crypto side, prior work documents short-horizon inefficiency and reversal with unconditional scaling statistics [1, 3, 16, 34, 50, 51], prices a weekly short-term reversal factor in the crypto cross-section [38], and describes a fragmented, retail-heavy structure [32, 41]; neither the existence of crypto mean reversion, nor its strengthening after large moves, nor the liquidity-provision reading is new on its own. What we contribute is the matched, population-scale, strictly out-of-sample cross-market measurement at intraday horizons, with its forward holdout and permutation-calibrated inference. The estimator's constrained-kernel form has a lineage in the econophysics literature on spin models of markets [8–10], set out in Appendix A; none of the empirical results depend on it.

2 Data, measurement device, and protocol

Data The primary universe is 15-minute candles over 2025-01-01 to 2026-02-11: the 183 highest-volume Binance USDT spot pairs and 187 liquid US stocks/ETFs (24-hour bars; regular-trading-hours restrictions where stated). Every dataset below is scored by the same models, walk-forward geometry and bootstrap as the primary universe, and none is reused to select any other's design. The supporting datasets are: a focal panel of fifteen instruments across six asset classes (crypto, equity indices, single stocks, gold, bonds, FX) at 5 m–4 h; multi-year 15-minute histories back to 2021-01; independent-venue refetches (Coinbase, OKX, Bybit) and quote-midprice bars (Dukascopy); the wrapper panel of Section 4 (Dukascopy spot metals and FX, Alpaca US-listed funds), with wrapper launch, BITO and uplisting eras back to 2021-10 and one-minute IBIT/Bitcoin bars for Section 4.2; Binance taker-flow fields and perpetual-futures series for Section 5; and the frozen universe refetched over 2026-02–2026-08 as the post-sample holdout. Each bar carries the intra-bar return $r_t = (\text{close}_t - \text{open}_t)/\text{open}_t$ and label $\text{up}_t = \mathbf{1}[r_t > 0]$; bars with $r_t = 0$ (*flat bars*) take label 0 under this convention, which is an artifact channel Section 3 treats explicitly. The convention pushes the median up-rate to 0.458 in crypto and 0.420 in stocks — the deficit is the flat-bar share almost exactly (median 6% and 16% of bars) — so one stock bar in six is labelled "down" by tie. Every headline statistic is therefore the base-rate-invariant AUC [24], raw accuracy is never a headline metric, and the flat-bar channel is carried as a separate effect size throughout. Construction, cleaning, and survivorship details are in Appendix G.

Measurement device The primary model is an autoregressive logit on the previous $N = 12$ soft-clipped returns $s_{t-k} = \tanh(\lambda r_{t-k})$, with lag weights tied to a decaying kernel:

$$P(\text{up}_t) = \text{sigmoid}\Big(C + A\sum_{k=1}^{N} k^{-\alpha}\, s_{t-k}\Big), \tag{1}$$

three free parameters (C, A, α) regardless of N. We call this the *constrained logit* throughout. The amplitude A carries the economics: $A < 0$ is mean reversion, $A > 0$ momentum, α the memory decay. The kernel tie is what makes it a low-variance instrument: L_1 and L_2 penalties shrink weights but treat every lag alike, whereas tying them to $A\,k^{-\alpha}$ forces the whole weight vector onto a two-dimensional monotone manifold, so effective degrees of freedom do not grow with N. Appendix A derives the restriction, notes its correspondence with a kinetic Ising chain, establishes what the data identify about the kernel (Appendix A.2: its decay, not its functional form), reports the one-lag baseline below the kernel (Table 5), and shows the three-parameter tie beating free, norm-penalized and tuned machine-learning baselines on out-of-sample log-loss. An unconstrained free AR(12) logit is fit alongside everywhere, and a result is *two-model significant* only when both clear the test, so "predictable" always means two different estimators agree. A third, parameter-free statistic accompanies them: the kernel-weighted sign correlation

$$R = \text{corr}\Big(\sigma_t,\ \textstyle\sum_{k=1}^{12} k^{-\alpha}\sigma_{t-k}\Big), \qquad \sigma_t = \text{sgn}(r_t), \quad \alpha \equiv 1, \tag{2}$$

with α fixed a priori and nothing fitted anywhere. This is the model-free counterpart used wherever a fitted result needs corroboration.

The primary test. *Universe:* 183 Binance USDT spot pairs (highest 24-hour quote volume at selection, excluding stablecoin-quote pairs and leveraged tokens) and 187 liquid US stocks/ETFs; exact symbol lists frozen in the replication package. *Data:* 15-minute candles, 2025-01-01 to 2026-02-11; label $\mathbf{1}[r_t > 0]$ on intra-bar open-to-close returns. *Models:* the kernel-constrained logit of Eq. (1) ($N = 12$, $\lambda = 150$) and the unconstrained free AR(12) logit, both by MLE with selection on a chronological validation tail. *Walk-forward:* train 5,760 / test 960 candles, step equal to the test block; non-overlapping out-of-sample blocks concatenated and scored once. *Statistic and inference:* out-of-sample AUC; one-sided moving-block-bootstrap p-value for AUC > 0.5 (block 384 candles, $B = 300$ resamples) per model, so the attainable p-floor is $1/(B+1) = 0.0033$; per-asset conjunction $p_\cap = \max(p_{\text{constr}}, p_{\text{free}})$; Benjamini–Hochberg FDR at $q = 0.05$ jointly across all 370 assets. *Population claim:* class-mean AUC gap under a joint moving-block bootstrap on the shared time grid ($B = 1{,}000$, raised to 5,000 in the sensitivity runs of Appendix B). *Exclusions:* assets with fewer than train + test + 2,000 candles, and assets with fewer than 2,000 out-of-sample candles; none other.

Protocol and inference Everything is scored strictly out of sample under the boxed protocol above, with per-asset moving-block-bootstrap inference [35, 45] and Benjamini–Hochberg FDR [5, 25] as boxed; *session-matched* regular-trading-hours (RTH) series, with 24-hour tapes and listed instruments alike restricted to the identical New York 09:30–16:00 slot grid, use a 60/10-trading-day walk-forward. The population statement is a *joint* moving-block bootstrap on the shared time grid, which preserves cross-sectional dependence; Section 3 states the estimand exactly and why per-asset tallies are descriptive rather than inferential. The primary test (universe, span, models, walk-forward geometry, statistic, FDR step) was fixed in design before the wide universe was run; the wrapper panel and the mechanism experiments were designed afterwards, on instruments outside its universe, and are reported as what they are. A refit-inclusive bootstrap, negative controls, and all protocol details are in Appendix B.

3 The population fact: reversal concentrates in crypto relative to US equities

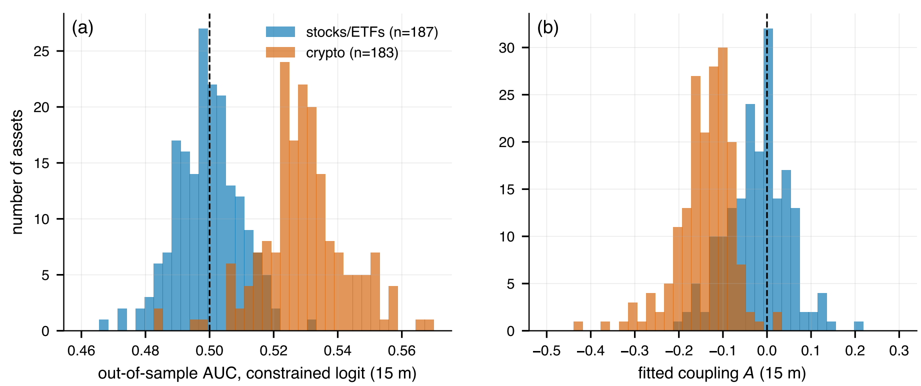


Figure 1: The primary test: 15-minute out-of-sample skill across 370 assets (183 Binance crypto pairs, 187 US stocks/ETFs), identical walk-forward and inference. (a) AUC by class; the separation shown is the all-bars scoring, and narrows under the artifact-robust rescorings of §3. (b) Fitted coupling A: mean-reverting ($A < 0$) for crypto, centered on zero for stocks.

Figure 1 is the population-level fact. Crypto AUC averages 0.531 with 98% of pairs above 0.5; stocks average 0.499 and split evenly around it, and after joint FDR control 90% of crypto pairs are significant against 2.7% of stocks (164 of 183 against 5 of 187). The joint moving-block bootstrap puts the class-mean gap at +0.031 (95% CI [+0.027, +0.035], $p < 10^{-3}$, stable across bootstrap block lengths; Appendix B), and the fitted coupling is mean-reverting in 98% of crypto pairs against a zero-centered stock distribution. The few crypto pairs without a signal are dominated by pegged stablecoins and brand-new microcaps; the few US-listed instruments clearing the raw test are led by the metal ETFs, which are listed claims on round-the-clock underlyings, a point Section 4 makes systematic. (Gold and silver miners also clear it on 24-hour bars, but with momentum-signed couplings and session-matched nulls, which is a different phenomenon; Section 4 uses them as controls.)

Because crypto pairs co-move so strongly, the estimand deserves an exact statement. It is the *difference in cross-sectional mean out-of-sample AUC between two selected universes*, the liquid Binance USDT cross-section against liquid US stocks/ETFs, with each asset's AUC computed once on its concatenated out-of-sample series over this sample window. Block resampling then treats that window as one draw from a stationary population of time blocks, which is the sense in which the interval below is a sampling statement. The "90% of 183 pairs" tally describes how broadly the signal covers that cross-section; it is not 183 independent replications, since with mean pairwise correlation 0.40 the effective number of independent crypto observations is roughly 2.5 (stocks: correlation 0.23, ≈4.2; the asymmetry is one more reason the gap is bootstrapped jointly rather than assembled from two class-level intervals). All inference therefore runs through the joint bootstrap, whose power comes from the time dimension. The defensible reading is "this liquid-crypto cross-section carried a small but pervasive and persistent reversal signal throughout the window," not "183 markets independently confirmed one." Small is the right word: 3.1 points of AUC before artifact adjustment, 2.2 after, 52.3% unthresholded accuracy on BTC. What earns the report is breadth and persistence, not size.

An ex-ante universe, reported in full The one researcher degree of freedom the fixed design could not remove is the universe itself: the 183 pairs were ranked by volume measured *after* the sample ended, so the as-fixed test runs on a survivor-tilted membership. We therefore report, with equal prominence, an **ex-ante specification** that removes the late-listing channel: the 133 of the 183 pairs already trading in the first sample month, re-ranked by that month's volume. The candidate pool is still the post-sample top 183, so this bounds the late-listing tilt rather than eliminating the volume ranking itself. Every headline statistic is unchanged or stronger on it. The class-mean gap is +0.035 (95% CI [+0.030, +0.038]) under the constrained logit and +0.030 [+0.026, +0.033] under the free logit; after joint FDR control, 97% of the 133 pairs are significant against the same 5 of 187 stocks (2.7%) as the primary specification, the equity leg being identical across the two since only the crypto universe varies; the coupling is negative in 98% (mean −0.15); and across ex-ante volume quintiles the AUC is flat, with a slightly *negative* volume–AUC rank correlation (−0.16). Late-listed survivors dilute rather than drive the effect, and predictability does not live in thin pairs. The selection rule still conditions on surviving to the download date (delisted pairs are unrecoverable from the public API), a caveat that applies to both universes and is disclosed as such; within what public data allow, the contrast does not depend on the part of the selection rule we can vary.

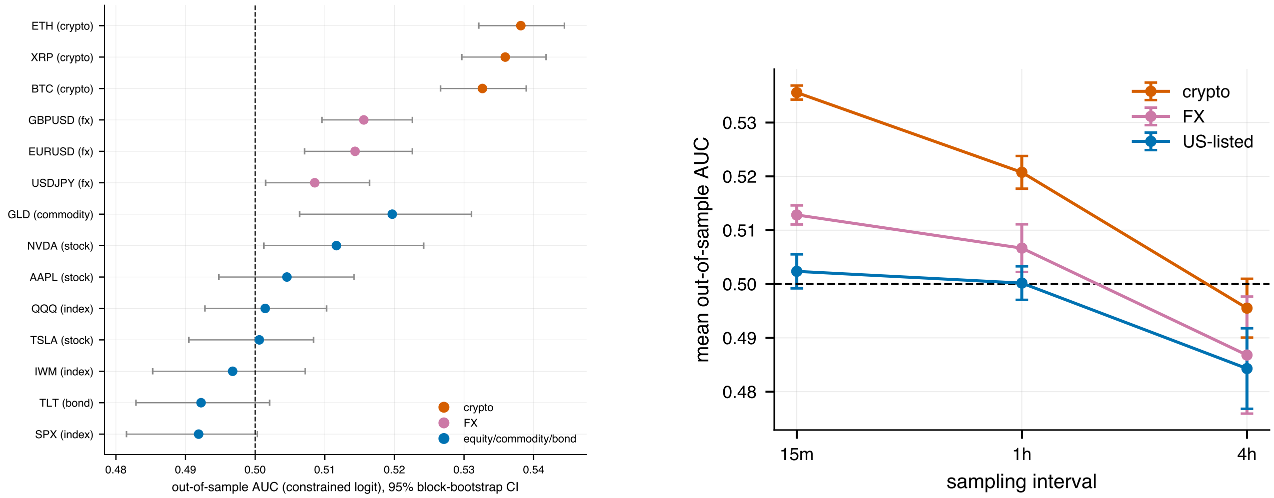
A six-month post-sample holdout The primary sample ends 2026-02-11, and every modelling choice in this paper (features, models, walk-forward geometry, bootstrap and FDR machinery) was frozen against data through that date; roughly six months later we refetched the identical frozen universe over 2026-02-12 to 2026-08-08 and reran the pipeline on outcome data that did not exist when any of them was fixed. Three sample-size thresholds were relaxed for the shorter span, and only those: minimum bars 8,720 to 6,960, minimum out-of-sample count 2,000 to 480, and the joint bootstrap's within-replicate minimum from 1,000 to 300 — the last is not a bar cut and bears on the interval below. The holdout FDR family is the 225 assets scored. No model, geometry, statistic or block length changed. The effect is attenuated but decisively present (Table 1): the joint dependence-preserving gap is +0.020 (95% CI [+0.010, +0.028], $p = 0.002$) under the constrained logit and +0.015 [+0.008, +0.021] under the free logit, against +0.031 in-sample, with 60% of scored crypto pairs still FDR-significant on roughly a third of the original sample length. The parameter-free sign statistic R of Eq. (2) is essentially *unchanged* on the crypto side (−0.034 against −0.035 in-sample, negative in 96%), while the stock side moves from −0.000 to −0.006 (66% negative), so the model-free class gap attenuates by a fifth against the fitted gap's third: the attenuation is mostly, not wholly, in the fitted model's out-of-period calibration. The ex-ante subset reads the same (mean AUC 0.521).

Three qualifications. Universe membership is not clean of the holdout window (the frozen list was ranked on volume as of 2026-06-08, a date inside it), so the holdout tests the overfitting objection, not the selection objection, which the ex-ante specification addresses. The model legs cover subsets (173 of 179 surviving crypto pairs, 52 liquid stocks) while the model-free statistic covers everything and is the leg to read for the full stock universe. And the attenuation decomposes: rescoring the *primary* window on exactly the holdout-scored assets gives +0.0287 against +0.0313 on the full universe, so of the total 0.0109 attenuation, 23% is the universe change and 77% is decay on a fixed universe (`holdout_did`) — the like-for-like comparison is +0.0287 in-sample against +0.0204 out. One six-month window cannot settle the adaptive-markets question, but the overfitting objection is tested directly and fails.

	Primary (2025-01–2026-02)	Holdout (2026-02–2026-08)
Class-mean gap, constrained	+0.031 [+0.027, +0.035]	+0.020 [+0.010, +0.028]
Class-mean gap, free logit	+0.026 [+0.023, +0.029]	+0.015 [+0.008, +0.021]
Model-free sign-gap (R, crypto − stocks)	−0.035	−0.028
Crypto mean AUC	0.531	0.522
Stock mean AUC	0.499	0.501
Crypto pairs FDR-significant	90%	60%
Crypto coupling $A < 0$	98%	96% (mean −0.21)

Table 1: The frozen pipeline on post-sample data. Gaps carry 95% joint moving-block-bootstrap CIs; the holdout model legs are the liquidity subsets described in the text, and the model-free row covers the full universes.

The focal panel (Figure 2a) places the two ends of the population contrast in a wider cross-section of market structures. Under the two-model criterion the tally is crypto 3/3, spot FX 2/3, US-listed 1/8: the three coins clear the no-skill line decisively, the dealer-intermediated European FX pairs show a faint copy, and exchange-listed equities, indices, and Treasuries show nothing. That ordering runs along intermediation of the primary market rather than along an equity/crypto dichotomy; the wrapper panel of Section 4 traces it further. (The one US-listed exception, GLD, is itself a wrapper on an OTC underlying, and its significance is carried by the hours outside the US session: on regular-trading-hours bars it reads null, like its underlying, which is the concordance Section 4 builds on.) Focal-panel detail: Appendix C.



(a) (b)

Figure 2: (a) The class ordering at instrument level: 15-minute out-of-sample AUC per focal instrument with 95% block-bootstrap CIs. Crypto clears 0.5; US-listed instruments cluster on it; spot FX sits between. (b) Mean out-of-sample AUC vs. sampling interval (±SE across assets, focal panel). The signal is strongest at the highest frequency and decays to no-skill within hours; the class ordering (crypto > FX > US-listed) holds at every horizon.

Finite memory, standing effect Two time dimensions frame what kind of object this is. Across sampling intervals (Figure 2b), the signal decays monotonically and is gone by four hours: the profile of a microstructure effect with finite memory rather than of slow-moving mispricing. At population scale the hourly rerun keeps 58% of the 169 crypto pairs meeting the hourly bar-count cut significant after FDR versus 0% of stocks. Across calendar time (Figure 3), it is a standing feature: every focal coin-year from 2021 through 2026 is two-model significant, the model-free sign correlation is negative in every calendar quarter, and the equity reference (SPY) scored identically over 2021–2025 never leaves the no-skill band. The contrast is therefore not an artifact of the 14-month primary window or of any single volatility regime. The mild downward drift in BTC's per-year AUC is the one feature consistent with slow adaptive erosion [31, 34, 37], and Section 6 states the corresponding falsification test.

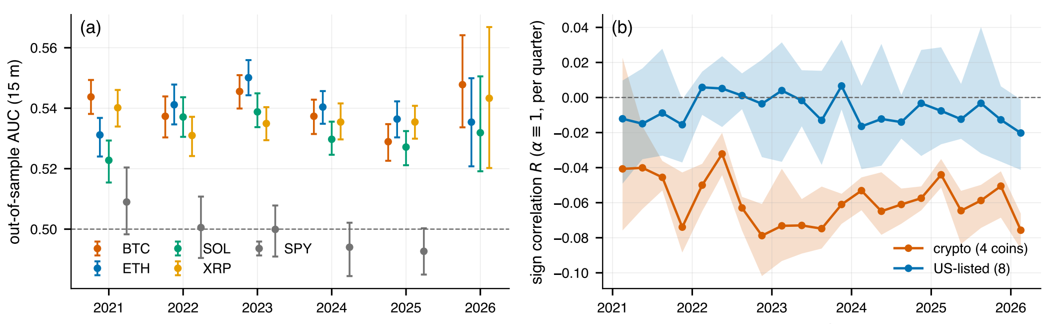


Figure 3: Five years of the same effect, read two ways. (a) Per-year out-of-sample AUC of the constrained model, four focal coins, 2021–2026 (95% bootstrap CIs; 2026 is a partial year), with SPY scored identically as the no-skill reference. (b) The model-free sign correlation R of Eq. (2) per calendar quarter (class mean; shading spans the min–max across assets): the crypto mean is negative in all 21 quarters, the US-listed mean stays near zero.

The contrast is a property of the data, not the estimator The lag-1 *return* autocorrelation is ≈0 on the focal coins, which makes the signal look invisible to the standard first check. The resolution (Figure 4) is that short-horizon crypto reversal is a *sign* phenomenon that strengthens with the size of the preceding move: the sign-flip rate rises monotonically across all ten deciles of $|r_{t-1}|$, from a base above 50% in every one (50.2 → 53.0%), and because ρ_1 weights bar pairs

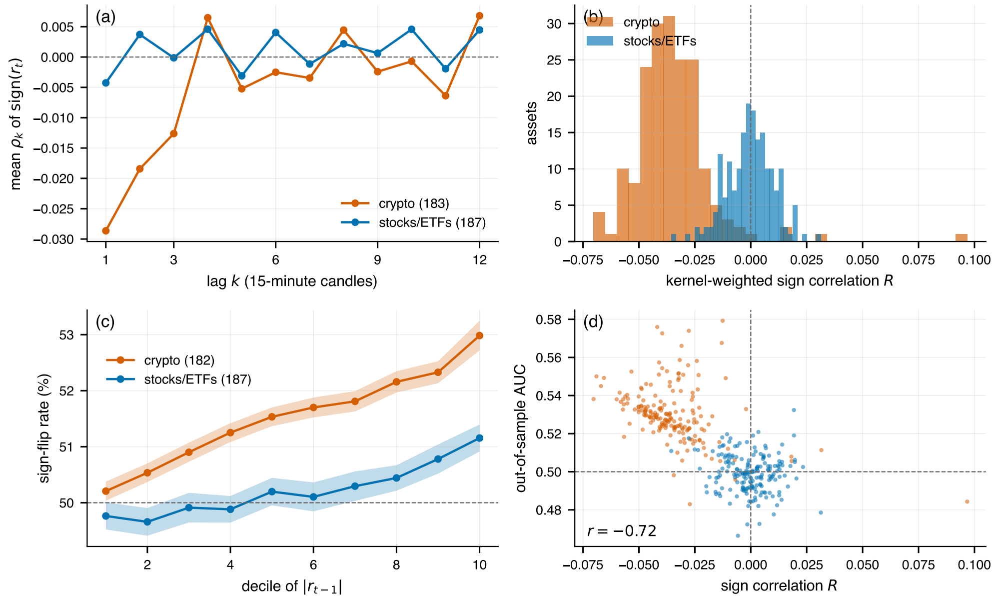


Figure 4: The same contrast with no fitted model. (a) Class-mean autocorrelation of $\mathrm{sgn}(r_t)$ by lag k. (b) Histograms of the kernel-weighted sign correlation R (Eq. (2), $\alpha \equiv 1$ fixed a priori), one count per asset. (c) Class-mean sign-flip rate by decile of the preceding move's size $|r_{t-1}|$, with shaded dispersion bands (one pegged pair with degenerate deciles is excluded from (c); 182 pairs). (d) R against the fitted model's out-of-sample AUC, one point per asset over all 370 ($r = -0.72$). Panels (a), (b) and (d) use all 183 crypto pairs.

by $r_{t-1}r_t$, the top decile's contribution nearly cancels the middle deciles'. Stocks share the gradient at half the slope and from a lower base (49.8 $\rightarrow$ 51.2%, non-monotone, at or below 50% through the fourth decile), so the size-dependence is not crypto-exclusive, only markedly larger there. Measured on signs over many lags the structure is not weak at all: the wide-universe class-mean lag-1 autocorrelations are -0.046 (returns) and -0.029 (signs) against -0.009 and -0.004 for stocks, the multi-lag sign autocorrelation is significant in 97% of crypto pairs versus 13% of stocks, and the parameter-free sign correlation R of Eq. (2) reproduces the class gap at -0.035 (95% CI $[-0.041, -0.029]$) under the identical joint bootstrap, negative in 98% of crypto pairs against 50% of stocks, with nothing fitted anywhere. The focal coins carry the same reading one instrument at a time (BTC/ETH/XRP/SOL sign ρ_1 -0.038, -0.056, -0.044, -0.038; Ljung–Box $Q(12)$ on signs $p < 10^{-15}$ throughout; R from -0.047 to -0.070), while their return ρ_1 is nowhere larger than 0.022 in magnitude. That is the sign–return divergence this paragraph opened with, in four instruments. Hurst exponents from detrended fluctuation analysis (DFA) corroborate at population level, with crypto antipersistent and a class-mean H gap of -0.049 (95% CI $[-0.084, -0.012]$) under the same joint bootstrap (Appendix D).

The same discipline applies to the kernel's own length: a one-lag special case, the probabilistic form of betting against the previous candle's sign, already reaches a crypto class-mean AUC of 0.527 against the twelve-lag kernel's 0.531. The population fact is therefore predominantly a lag-one reversal with a short memory tail; Appendix A.3 quantifies the kernel's small, systematic increment and its saturation by $N \approx 3$.

The fact survives every artifact channel Table 2 compresses the robustness battery; three rows carry the most weight. Restricting the *entire* stock universe to regular trading hours, so thin overnight bars can neither mask nor manufacture structure, leaves stocks at mean AUC 0.498 while 24/7 crypto is unchanged. Dropping flat bars removes the one channel that rewards a reversal model with free "correct" calls, since up = $\mathbf{1}[r > 0]$ scores an unchanged price as a down move; it moves wide-crypto AUC from 0.531 to 0.520 with the stock side unmoved. Skipping one bar between features and label removes pure boundary effects such as bid–ask bounce [46], and retains 71% of the crypto excess with 81% of pairs still significant. The remaining rows close off staleness, single-venue idiosyncrasy, and estimator dependence; details are in Appendix D.

The paper carries two effect sizes, and they stay separate throughout. The **primary analysis**, whose design was fixed ex ante (§2), is the all-bars gap of +0.031 (95% CI [+0.027, +0.035]). The non-flat rescoring is our **preferred artifact-robust effect-size estimate**: crypto 0.5200 against stocks 0.4985, a class-mean gap of +0.022 (95% CI [+0.019, +0.025]) under the same joint bootstrap, and +0.017 [+0.015, +0.019] under the free logit. The latter is a smaller number reached by a decision made after seeing the data, so never a substitute for the primary figure, and its interval is well clear of zero and of the primary interval's lower end alike.

Artifact channel	Test	Result
Trading-session mismatch	all 187 stocks on RTH-only bars	gap intact (stocks 0.498)
Flat-bar labels	rescore on non-flat bars only	gap +0.022 [+0.019, +0.025]
One-bar bounce / staleness	one-bar gap between features and label	71% of excess survives
Thin-pair staleness	AUC vs. volume cross-section	volume slope ≈ 0; flat quintiles
Single-venue idiosyncrasy	refetch from Coinbase, OKX, Bybit	12/12 cells within ±0.002
Bid–ask bounce	quote-midprice bars (bid/ask average)	unchanged (BTC 0.529, ETH 0.537)
Return convention	close-to-close, VWAP variants	identical to third decimal
Quarter-hour clock effects	bars rebuilt at +1/+2/+5/+7 min offsets (focal coins, 1 s data)	R strongly negative at every offset
Pipeline manufacture	shuffled labels, all 370 assets	0/370 significant after FDR
Magnitude structure	sign-randomized surrogates, $\lvert r \rvert$ path fixed	0/14 significant; crypto mean 0.496
Regime dependence	per-year walk-forward 2021–2026	24/24 coin-years significant
Sample-tuned choices	frozen pipeline on post-sample 2026-02–2026-08 data	gap +0.020 [+0.010, +0.028]
Estimator dependence	free logit; sign statistics; DFA	gap +0.026; −0.035; −0.049 [−0.084, −0.012]
Kernel length	one-lag ($N = 1$) special case	gap +0.029; +0.004 crypto gain to N=12
Under-powered equity null	per-stock power; joint bound on class mean	median power 0.98; mean ≤ 0.5007
Both channels at once	non-flat *and* one-bar gap, jointly	gap +0.011 [+0.008, +0.014]
Inference machinery	exact permutation null, day-aligned label shift	0/340 shifts reach the gap; 4.4 null SD
Intraday-variance detectability	AUC within hour-of-day strata	gap *widens* to +0.034
One common factor	equal-weight crypto factor projected out	81% of R retained; 98% still negative

Table 2: The artifact ledger. Each row is a mechanical or statistical channel that could manufacture the crypto–equity contrast, the test that isolates it, and the outcome. Full details, including the phase-randomized surrogate control and its anti-conservativeness diagnosis, are in Appendices D and B.4.

Both artifact channels at once One-at-a-time robustness overstates what survives, so we also report the conjunction: the walk-forward refit with a one-bar feature/label gap, scored with flat bars dropped, under the identical joint bootstrap (`joint_artifact`; the gap-only cell reproduces the published single-channel figures exactly, which makes the joint cell comparable). Both adjustments together leave crypto at 0.5093 against stocks 0.4980: a class-mean gap of **+0.011** (95% CI [+0.008, +0.014]). That is 36% of the headline gap, below the 47% independent channels would leave, so the two channels overlap — part of what the gap removes is structure the flat-bar convention was manufacturing. The paper's effect size is therefore a range, every point of it clear of zero: +0.031 as fixed ex ante, +0.022 on either adjustment alone, +0.011 with both.

The equity null is powered Absence of evidence licenses evidence of absence only with a power calculation, so we state one. Using each stock's own moving-block-bootstrap AUC dispersion (the identical block geometry as the per-asset test), the one-sided 5% test would detect a true effect of the crypto class-mean size (+0.031) with median power 0.98 across the 187 stocks; power exceeds 0.8 in 94% of them, with only the shortest series falling to ≈0.64. The class-level statement is sharper: the joint moving-block bootstrap puts a one-sided 95% upper bound of **0.5007** on the stock class-mean AUC, so the design does not merely fail to find an equity-class effect; it bounds any common one at less than a fortieth of the crypto class mean's excess (Appendix B.3).

Two readings the design must exclude, and does First, *detectability*. AUC ranks bars by a score that is a fixed function of raw returns, so a steep intraday variance profile can bury a real dependence: intraday heteroskedasticity is 17.6× for listed instruments against 5.0× for crypto, and SPX — whose lag-1 return autocorrelation of −0.030 is ten times BTC's — reads AUC 0.492 yet lifts to 0.515 when its variance is homogenized (Appendix B.4), essentially the elliptical-implied linear reading $\Phi(\sqrt{4/\pi}\,\lvert\rho\rvert) = 0.5135$. Could the whole class gap be a detectability gap? No: recomputing each asset's AUC within hour-of-day buckets, which removes the cross-regime pooling exactly and needs no refit, *widens* the gap from +0.0313 to +0.0335 (`factor_variance`). The check holds the fitted score fixed, and the equity null remains a statement about a common effect measured this way, not proof that US equities carry no linear reversal at all; the crypto side was never a candidate, since BTC's ρ_1 of −0.003 supports only 0.501 through the linear channel against 0.533 observed.

Second, *one factor wearing 183 names*. With $N_{\text{eff}} \approx 2.5$ the cross-section could be a single mean-reverting market factor. Projecting an equal-weight crypto factor out of every pair (betas estimated on the first half of the span only) and recomputing the parameter-free R of Eq. (2) on the residuals: the class mean moves from −0.0358 to −0.0290, retaining 81%, and stays negative in 98% of pairs (`factor_variance`). Four fifths of the sign reversal is idiosyncratic. The residuals are not independent either — inference still runs through the joint bootstrap and the permutation null of Appendix B.2 — but the one-factor reading is ruled out.

4 The signature in listed wrappers

Does the reversal belong to the asset's price process or to the venue it trades on? A cross-market contrast cannot say, since it varies both at once. Listed wrappers vary them separately: eight underlyings (four spot metals on OTC dealer networks, Bitcoin and Ether on 24/7 exchanges, EURUSD and USDJPY on 24/5 spot FX) each have NAV-linked US-listed funds — seventeen spot-custody wrappers plus the futures-based BITO and the 2× leveraged BITU and ETHU — trading with designated market makers, regular hours, and a consolidated tape. Five listed instruments *correlated with* but not priced off an underlying (GDX, NEM, COIN, MSTR, UUP) separate "NAV-linked" from "same risk factor." Every leg, including the 24/7 and OTC underlyings, is scored on the identical New York 09:30–16:00 15-minute slots under the identical walk-forward, models, and bootstrap. Figure 5 shows the panel; Table 3 aggregates it to the family level, where observations are independent.

One limit governs how the whole section should be read, and we state it before the results. IBIT and Bitcoin move together at $r = 0.982$ per minute (§4.3), so at 15-minute resolution a liquid wrapper's bars nearly *are* its underlying's bars, and inheritance on the live cells is close to mechanical: under arbitrage it is the maintained hypothesis, not a discovery. What the panel adds is descriptive but useful — the correlated controls carry nothing, the nulls are concordant, thin tapes attenuate — and one practical corollary with teeth: efficiency statistics computed on a wrapper's tape describe the wrapped process, not the listing venue, so listed-universe comparisons that pool wrappers with native instruments are misspecified (Section 6).

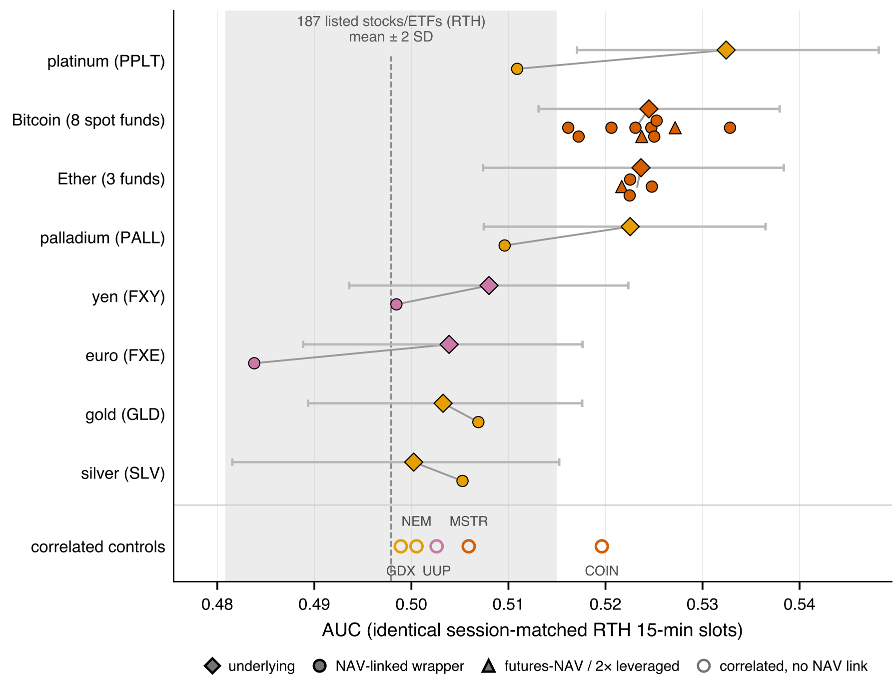


Figure 5: The wrapper panel on session-matched RTH bars. Diamonds: underlyings with 95% bootstrap CIs; filled circles: their NAV-linked wrappers (triangles: futures-NAV and leveraged funds); grey connectors span the family gap; open circles: correlated controls; shaded band: 187-stock RTH reference (mean ±2 SD). Live underlyings (Bitcoin, Ether) are matched by their funds right of the band; null underlyings (gold, silver, FX) sit in-band with theirs; the thin metal wrappers fall back toward the band. Values: Table 3, Appendix E.

4.1 Panel results

Where the underlying carries a signal on the session-matched slots, its NAV-linked funds carry it too: the Bitcoin and Ether families match their underlyings to a family gap of 0.001 or less, three cross-sectional SDs above the 187-stock reference. Where the underlying reads a null on those slots — gold and silver, whose marginal 24-hour readings do not survive the session restriction — its funds read the null. Instruments carrying the same risk without the NAV link carry nothing: 0 of 5 controls, including COIN and MSTR, whose Bitcoin betas dwarf any metal ETF's tracking error. Thin tapes attenuate: PPLT reads 0.511 against platinum's 0.532 and PALL 0.510 against palladium's 0.523 (16–18% flat bars on their 24-hour tapes; neither cell individually significant, conj. $p = 0.129, 0.157$), consistent with stale prints offsetting inherited reversal, though without intraday NAV the panel cannot separate stale sampling from partial non-transmission. The omitted thinner Bitcoin and Ether funds could only widen the measured family gaps, so the panel's composition

Underlying family	Wrappers	Underl. AUC	Wrapper mean	Family gap	SD units vs. stocks
Bitcoin	8 spot funds	**0.524**	**0.523**	−0.001	+3.0
Ether	3 funds	0.524	0.523	−0.000	+3.0
Gold	GLD	0.503	0.507	+0.004	+1.1
Silver	SLV	0.500	0.505	+0.005	+0.9
Platinum	PPLT (thin)	**0.532**	0.511	−0.022	+1.5
Palladium	PALL (thin)	**0.523**	0.510	−0.013	+1.4
Euro	FXE (thin)	0.504	0.484	−0.020	−1.7
Yen	FXY (thin)	0.508	0.499	−0.010	+0.1
correlated controls	GDX, NEM, COIN, MSTR, UUP		0/5 significant; mean AUC 0.506		

Table 3: The panel at the family level. Funds on one underlying share one price process and window, so they are within-family replications, not separate tests; family gaps are computed at full precision. The last column standardizes the wrapper mean against the 187-stock RTH cross-section (mean 0.498, SD 0.0085) — a descriptive scale, not a test statistic; the control COIN sits at +2.6 on it. Per-cell inference: Table 7. Bold: two-model significant at 5%; "thin": > 15% flat bars.

flatters no conclusion.

Formal inference is weaker than the pattern, in three specific ways. First, the eight Bitcoin funds share one process and one window: eight within-family replications (SD 0.005, range 0.516–0.533), not eight confirmations. Applying the wide study's own FDR discipline within the 33-leg panel family, 9 of 13 nominally significant legs survive; the four that fail are BTCO, EZBC, BITO — and Bitcoin itself — so the count is 6 of 8 spot-Bitcoin funds, and the futures-NAV cell does not clear the corrected bar (`panel_fdr`). Second, regressing the family wrapper mean on its underlying gives an inheritance slope of 0.69 (SE 0.32): it rejects neither full inheritance ($p = 0.37$) nor *zero* inheritance ($p = 0.076$), so eight families cannot separate the hypotheses the panel was built around, and we retire the permutation p-values we previously reported. Third, COIN — kept on the control side because it fails free-logit corroboration ($p = 0.36$; constrained $p = 0.001$) — sits at +2.6 on the panel's own descriptive scale, above every wrapper family except Bitcoin and Ether. Read strictly, the panel separates live-underlying NAV-linked wrappers from everything else, not NAV-linkage from correlation as such. Ether's cells are likewise concordance rather than per-cell significance. What stands is the direction of the whole pattern; we offer no formal test of it.

4.2 Transmission in time

A cross-sectional concordance still admits a slow story in which listed venues gradually learn the underlying's dynamics. Three event designs say otherwise (Figure 6, Table 8).

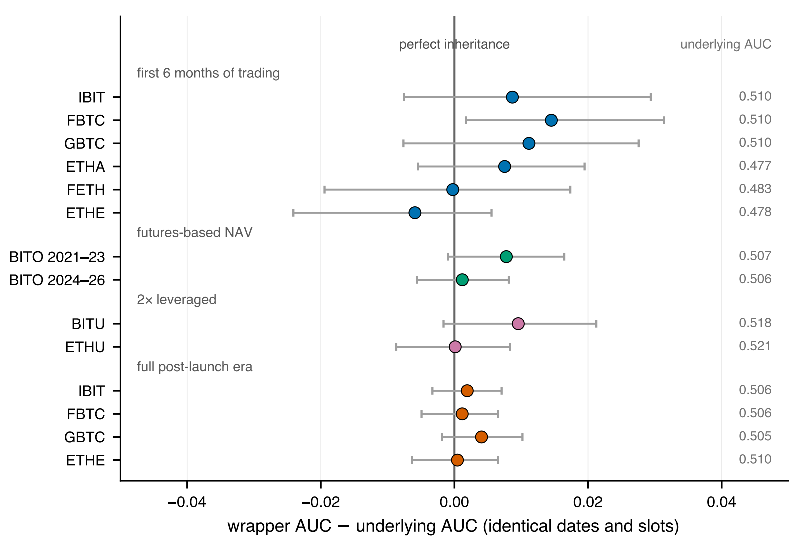


Figure 6: Wrapper AUC minus its underlying's on exactly shared bars, with 95% paired moving-block bootstrap intervals (one block draw applied to both legs). Zero is perfect inheritance; the right-hand column gives the underlying's AUC on the shared bars. Per-leg readings: Table 8.

Bitcoin's three 2024 funds carried the coupling in their first six months of exchange trading, at fitted values matching their underlying's (A_w between −0.07 and −0.12 against −0.08); the cells are individually insignificant at ~1,700 bars, so they establish concordance, not detection. BITO, priced off CME futures and predating every spot ETF, matches its underlying over 12,813 bars and is the one era cell that is itself two-model significant at raw 5% (neither this era cell nor BITO's panel-span leg survives an FDR correction). The 2× leveraged BITU and ETHU reproduce their families' readings through swaps and cash-settled futures, and the three Ether funds, launching into a window where Ether itself read below 0.5, reproduce approximately that rather than a positive signal no process could have handed them.

Formally: thirteen of the fourteen gaps are indistinguishable from zero, a non-rejection that short launch windows make undemanding; the stronger statement is that 13 of 14 intervals *exclude* complete non-inheritance (wrapper AUC = 0.5), FETH's launch semester being the exception. Two residuals cut the other way. Twelve of fourteen gaps are positive (mean +0.004; two-sided sign-test $p = 0.013$): wrapper tapes read slightly above the process they wrap, an offset we cannot attribute — venue-added structure and wrapper-side tick bounce would both produce it — and the panel cannot exclude. And FBTC's launch interval excludes zero (+0.015, CI [+0.002, +0.031]), about the one exceedance in fourteen a 5% interval produces, in the offset's direction.

4.3 One-minute timing

Over six months of aligned RTH minutes ($n = 49{,}170$), IBIT and Binance BTC move together at $r = 0.982$, while the largest of the twenty displaced correlations at ±1 to ±10 minutes is 0.013 — about one five-thousandth of the contemporaneous covariation in R^2. There is no exploitable spot→wrapper lead at this resolution, as expected if arbitrage enforces the NAV link inside the sampling minute. The displaced terms are not noise: they alternate in sign lag by lag and track Bitcoin's own one-minute autocorrelation, i.e. the underlying's weak reversal read through the contemporaneous link. Summed by direction, the wrapper→spot side is the stronger (+0.015 against −0.005; difference CI $[-0.028, -0.012]$, excluding zero), driven mainly by the −0.010 lag-one spot→wrapper term. We read the asymmetry as IBIT's transient pricing error mean-reverting within the minute, an interpretation we have not tested; settling it needs intraday NAV. Its economic content is bounded either way: the displaced terms sum to under 0.1% of incremental R^2 for the next Bitcoin minute, far inside any cross-venue round-trip cost. Construction and diagnostics (alignment placebo, staleness screens, IEX-feed caveats): Appendix E.1.

The contemporaneity sets the panel's resolution limit. Skipping one bar between predictors and label severs the contemporaneous echo and leaves nothing significant anywhere on the session-matched slots (underlying 0.507, largest wrapper 0.497); on the underlying's five-times-longer 24/7 tape the gapped effect survives (BTC 0.514, $p < 10^{-3}$), but that comparison changes the bar set as well as the instrument. A wrapper's bars carry the underlying's reversal because each bar nearly *is* the underlying's bar; what survives across the wrapper's own bar boundary once the link is cut is below this design's resolution, so we make no claim about a venue margin. Which arbitrage channel binds is not identified and need not be: all of them transmit the process, and the timing rules out only the stories in which the listing venue's own trading generates the signal on a timescale its bars could resolve.

5 Mechanism evidence: reversal, aggressive order flow, and capture costs

Back on the originating tape, two questions remain: what the reversal travels with, and why nothing has removed it.

5.1 Reversal follows aggressive flow

Binance klines record each bar's taker-buy volume, signing the aggressor side of every trade without order-book data: the bar's taker imbalance is $i_t = 2V_t^{\text{taker-buy}}/V_t - 1 \in [-1, 1]$. Call a bar *flow-driven* when its price move agrees in sign with its imbalance (aggressors pushed the price the way it went; ~70% of bars) and *flow-opposed* otherwise. A liquidity-provision account [21, 42], in which prices concede to absorb incoming flow and then recover as inventories are worked off, predicts reversal concentrated after flow-driven moves and increasing in the flow. This is the classic volume- and imbalance-conditioned reversal of equities [11, 12], measured here on a market whose tape signs the aggressor side of every trade directly.

Both predictions hold on all four focal coins (Figure 7ab), and the walk-forward model agrees: its out-of-sample AUC is 0.540 after flow-driven bars versus 0.519 after flow-opposed ones, so the taker tape tells you when the coupling will work. The association also holds across the universe (Figure 7c), where the flip-rate difference retains an independent loading on out-of-sample AUC when volume and trade size enter jointly, while a simple retail proxy, the median dollar trade size, carries nothing of its own (Appendix F). What lines up with the coupling is how hard flow-pushed moves revert, not how small the customers are.

Flow conditions the reversal but does not replace it (Table 4): adding twelve lags of imbalance to the sign path adds nothing out of sample, while the sign path beats flow alone decisively. At bar granularity the imbalance is informationally redundant with the price path it helped create — the signature of overreaction-plus-correction rather than of a signal that merely proxies inventory pressure. The pieces map onto the classic liquidity-provision mechanism [21, 42]: demanders of immediacy push price through the book, the concession scales with how hard they push, and the price recovers as suppliers work off the position, while a move against the aggressor flow required no concession and reverts far more

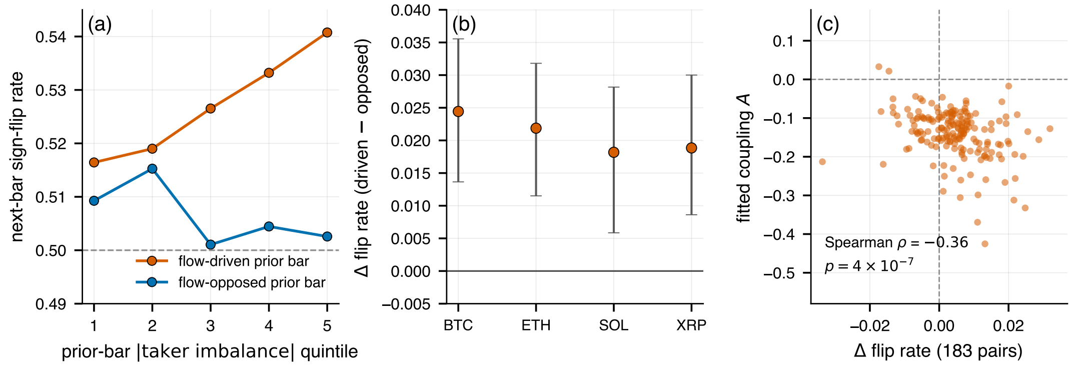


Figure 7: Reversal concentrates after flow-driven moves and grows with the flow. (a) Pooled next-bar sign-flip rate by quintile of the prior bar's |taker imbalance| (four focal coins, 15 m, ~39k bars each), split by whether the prior move agreed with its flow: a dose–response after flow-driven bars, a coin-flip after flow-opposed ones. (b) The flow-driven minus flow-opposed flip-rate difference per coin with moving-block bootstrap CIs (pooled: +0.021, CI [0.015, 0.026]). (c) The conditioning scales to the cross-section: across the 183 pairs, those whose flow-driven moves revert harder carry systematically stronger fitted reversal couplings (three extreme-$|A|$ pegged/microcap pairs outside the axis range).

weakly (AUC 0.519 against 0.540; an attenuation, not a null) — the transitory price-pressure pattern that Hendershott and Menkveld [27] estimate structurally in equities, read here at bar level. Three limits bound this reading. Bar-level aggressor volume is not an order book, so depth-consuming and quote-driven moves are not separated; the "flow-driven" label is partly constructed from the contemporaneous return's own sign, so the split is a conditioning, not a treatment; and the time-reversal control of Appendix B.4 shows most of the detectable sign structure is time-symmetric, so the directional overreaction-and-correction story rests on the flow conditioning here, not on the sign path itself. Conditioning on the perpetual basis, funding rate, funding clock, volatility and session is flat in every case (Appendix F).

Walk-forward feature set	OOS AUC	paired ΔAUC vs. signs (CI)
12 sign lags	0.529	—
12 taker-imbalance lags	0.521	−0.008 [−0.012, −0.005]
signs + imbalance	0.529	0.000 [−0.002, +0.002]

Table 4: The subsumption test, pooled across the four focal coins: bar-level flow neither adds to nor replaces the sign path. Coefficient detail: Appendix F.

5.2 The edge is priced inside benchmark transaction costs

Why, then, has a measurable signal survived? This subsection converts the statistical edge into economic units and prices it against what capturing it would cost (Figure 8). A deployed rule acts only when model confidence clears a threshold $|p_t - \frac{1}{2}| \geq \tau$, with τ fixed on an ex-ante grid, so nothing is selected on the out-of-sample series. Raising the threshold does exactly what a real signal should do to directional accuracy: it rises monotonically as coverage falls over the range where the trade count supports an estimate, clearing 56% on the most confident bars, and rises more steeply for the constrained model than for the free logit, so model confidence genuinely ranks accuracy rather than merely scaling it.

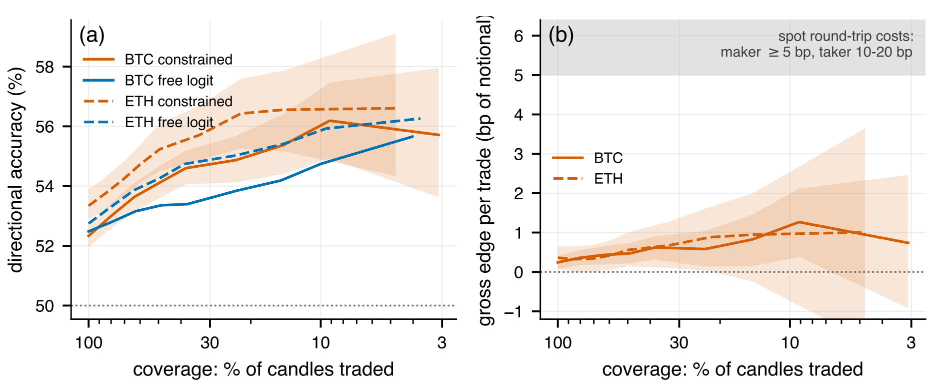


Figure 8: The cost of capture (BTC/ETH, 15 m). (a) Directional accuracy as the confidence threshold restricts coverage. (b) The same thresholds in return units: the gross edge per trade stays below the shaded spot round-trip cost band throughout, peaking near 1.3 bp against a 5 bp cheapest band; the 10–20 bp taker band lies above the axis range. Bands: 95% bootstrap CIs.

The same thresholds in return units tell the economically decisive half of the story: the gross edge per trade never leaves the low single digits of a basis point. At its most selective it is about a quarter of the cheapest realistic spot round-trip band and a tenth of the taker band, and unthresholded a twentieth of the former. Thresholding cannot rescue it, because trading fewer, better bars shrinks the opportunity count faster than it grows the edge. This is a population statement, not a focal-coin one: not one of the 183 crypto pairs clears even the 5 bp maker band at any threshold, the median pair earns 0.46 bp per trade at $\tau = 0.02$ where it trades at all against zero for the median stock, and moving to 5-minute bars makes matters worse rather than better (0.15 bp), because higher frequency shrinks each opportunity faster than it shrinks costs. On BTC-15 m the grid runs 52.3% accuracy unthresholded to 54.6% at $\tau = 0.02$ (38% of bars traded) and 56.2% at $\tau = 0.05$ (9%), the mean gross edge rising 0.24 bp to ≈1.3 bp over the same range; past $\tau = 0.05$ the trade count collapses (3% of bars at $\tau = 0.075$, 0.9% at $\tau = 0.1$) and the grid is uninformative in either direction. Accuracy comparisons use the focal coins only, since flat bars inflate accuracy on thin instruments. Explicit fees, not the quoted spread, are what bind here: top-of-book on the focal Binance pairs is of order a tick, well inside the edge, so a maker who never crosses still pays the fee schedule (Appendix G).

The precise claim is that the edge is **not exploitable under benchmark spot-cost assumptions**: a public-schedule taker or maker paying explicit spot costs cannot capture it anywhere in the cross-section. That is the statement the limits-to-arbitrage reading [22, 48] needs, and it closes the loop with the flow evidence above: a compensation for supplying immediacy, competed down to roughly the cost of competing for it, is exactly the residual a costly liquidity-supply equilibrium leaves standing. It is not a statement that no trading technology could reach the edge. Sophisticated participants face a cost object we cannot price (fee tiers and rebates, passive fills, futures, inventory netting), and to whatever extent they do capture part of the flow-provision compensation, they *are* the liquidity suppliers of Section 5.1, which reinforces the account rather than threatening it. Which cost component binds hardest this evidence cannot rank; the gross figures additionally ignore depth, latency, and adverse selection, all of which move against a would-be taker of the signal.

6 Discussion

The account At the evidential weights of Section 1: short-horizon mean reversion is strongest in price processes whose primary discovery occurs on thin, round-the-clock, lightly intermediated tapes; a claim listed on such a process carries its signature; and it persists at a size not exploitable under benchmark spot-cost assumptions [22, 42, 48]. The apparent US-listed "exceptions" of Section 3, the metal ETFs, are then not exceptions but wrappers on elsewhere-discovered processes; the miners, which clear the 24-hour test with momentum-signed couplings and read null on session-matched slots, are a different phenomenon and serve as controls without contradiction.

Implications for measuring efficiency Two practical corollaries follow, both testable on existing data as new wrapper families launch. First, an instrument's listing venue is the wrong unit for efficiency measurement whenever the instrument is priced off an external process. Cross-sectional comparisons that mix native-discovery instruments with wrappers, as most listed universes do, will misattribute the wrappers' readings to the listing venue unless the NAV linkage is modeled. Second, the composition of measured "anomalies" in listed markets should track what gets wrapped: as ETFs extend into more round-the-clock or lightly intermediated underlyings, their tapes will import those processes' short-horizon signatures at full size, without any change in the listing venue's own efficiency.

The equity mirror The account also rationalizes the stock side of Figure 1: short-horizon reversal was once large in US equities and shrank as intermediation industrialized [13, 14, 29, 36, 42]. On our 15-minute bars that history survives as a small negative coupling in single stocks (AAPL, NVDA $A \approx -0.13$; Appendix C) that no longer suffices to rank direction out of sample. Crypto's fragmented, retail-heavy tape [32, 41] sits where equities sat before liquidity provision scaled, and the wrapper evidence shows that listing a claim on that process inside modern equity structure imports the signature rather than curing it.

Limitations and falsification The wide crypto cross-section is single-exchange and 14 months long, and even the ex-ante specification conditions on pairs surviving to the download date (§3); the focal coins replicate across venues and years. The six-month holdout is one window: it confirms survival, not stationarity, and extending it as data accrue costs nothing but time. The wrapper panel rests on eight underlying families in one 14-month window, only two of them (Bitcoin, platinum–palladium jointly) with strongly live underlyings on regular-hours slots; its per-cell, permutation, and resolution limits are stated where they arise (§4.1, §4.3), and the trust eras of GBTC and ETHE predate our feed, so conversion events reduce to inheritance-from-uplisting. The mechanism and cost evidence are bounded as stated in Section 5: bar-level aggressor volume rather than book states, signatures rather than structural identification, and gross benchmark costs that do not rank which cost component binds.

Bounded is not unfalsifiable, and each leg is exposed. A NAV-linked wrapper on a live underlying that fails to inherit, or a correlated control that robustly does; a wrapper leading its underlying at one minute by enough to matter (the asymmetry we find sums to under 0.1% of incremental R^2 and would have to grow an order of magnitude, and survive an intraday-NAV decomposition, to bear on this); reversal concentrating after flow-*opposed* moves in another market; a venue or period in which the gross edge exceeds its own round-trip cost band; walk-forward AUC on future data converging to 0.5 — the adaptive-markets reading [34, 37], which the mild downward BTC drift and the attenuated holdout keep live, the first six months of forward data rejecting convergence but not erosion; reversal vanishing when the 15-minute grid is rebuilt at off-grid offsets (on the focal coins it does not, Appendix D); or order-book data showing reversal absent after depth-consuming moves.

7 Conclusion

Short-horizon directional mean reversion is concentrated in cryptocurrency markets relative to US equities: a population fact, measured under one matched out-of-sample protocol, that needs no fitted model to see and survives every artifact channel we can test, an exact permutation null, the removal of the common crypto factor, and a frozen six-month holdout. Its size is a range, not a point — three AUC points as designed, one with both artifact adjustments applied at once — and every end of the range is clear of zero. Listed wrappers on such processes carry their underlying's reading, nulls included, which we read descriptively: a wrapper's tape describes the process it wraps, not the venue it prints on. What sustains the reversal we bound rather than identify: it concentrates after aggressive-flow-driven moves and scales with flow intensity, as compensated liquidity provision would, and it survives at an edge not exploitable under benchmark spot costs. The sharpest open tests need data we lack at scale: order-book depth on the originating tape, and intraday net asset value for the thin wrappers.

Funding

This research received no funding from any agency in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author operates an automated trading system for cryptocurrency binary (prediction-market) contracts that uses a variant of the model studied here, on a different instrument and against a different cost object from the spot round-trips priced in Section 5.2. The 15-minute horizon and the binary up/down label mirror the contract that system trades; they predate this study and were not tuned on its results. No data from that system are used in this paper, and no parameter, threshold, or design choice reported here was selected on its results; the fixed constants (λ, N, the kernel family) are shown inert in Appendix D. The system's existence bears on Section 5.2 in one way worth stating: fixed-odds binary contracts carry a cost object with no per-notional spot fee, which is consistent with the boundary drawn there — the non-exploitability claim is conditional on spot cost structures, not on every venue that prices the same 15-minute direction. No other competing interests.

Data availability

All market data are obtainable from public sources (Binance, Coinbase, OKX, Bybit and Dukascopy public APIs; Alpaca market-data API); a complete replication package is at `https://github.com/nadav2/short-horizon-reversion` (Appendix H).

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A The measurement device

A.1 Spin transform and model

Asset returns are heavy-tailed [15, 16], so a single outlier bar can dominate a linear field. Each return is mapped to a *soft spin* $s_t = \tanh(\lambda r_t) \in (-1, +1)$ with $\lambda = 150$ fixed (not fitted); the decision boundary is insensitive to λ over $[50, 1000]$ (Appendix D). Because λ is fixed while volatilities differ across assets, cross-class comparisons of coupling *magnitudes* use a volatility-standardized variant $s_t = \tanh(c\, r_t/\hat{\sigma})$, $c = 0.45$, with $\hat{\sigma}$ estimated causally on the training window; every cross-market conclusion is unchanged under it. The working model is the kernel-tied logit of Eq. (1): an AR(N) logistic regression whose lag weights are constrained to $w_k = A\, k^{-\alpha}$, leaving three free parameters (C, A, α) however many lags enter.

Kinetic-Ising correspondence Equation (1) has the form of the one-step Glauber transition law [19] of a kinetic Ising chain with $J_k = J_0 k^{-\alpha}$ couplings [17], under $C \equiv 2\beta h_0$, $A \equiv 2\beta J_0$ (β not separately identifiable). The correspondence is exact only as $\lambda \to \infty$, where $s_t \to \mathrm{sgn}(r_t)$; at finite λ the model conditions on $s_t \in (-1, 1)$, i.e. the naive mean-field closure with s_t a local magnetization. At $\lambda = 150$ the transform is near-linear over the bulk of 15-minute returns and saturates only above ~1%, so s_t is in practice a soft-winsorized return, only the product $A\lambda$ is identified in the bulk, and the $\lambda \in [50, 1000]$ sweep of Appendix D is a tail-clipping check, not evidence about λ. Unlike Dyson's equilibrium chain the couplings are one-sided in time and antiferromagnetic — no detailed balance, no Boltzmann equilibrium; we borrow the coupling form and nothing else. The multivariate mapping of Campajola et al. [10] requires non-negative couplings and excludes our regime, but the single-chain correspondence holds for either sign of A. The magnetic vocabulary is retained purely as labels, and no empirical claim depends on it.

A.2 Kernel-shape identifiability

Is the power-law form a finding or a choice? Refitting with three tied families under the identical protocol (power-law $A\, k^{-\alpha}$, exponential $A\, e^{-(k-1)/\tau}$, both three parameters, and flat A with two), the *decay* is identified: the flat kernel is worse in 13/15 within-crypto cells, significantly so in 9 (paired block bootstrap), with large 15 m deficits (ETH 0.527 vs. 0.544). The functional form is not: power-law and exponential differ by less than $|0.003|$ AUC in 14/15 cells with sign flips across cells. We therefore interpret only the existence, sign, and rough decay of the memory kernel; the fitted $\alpha \approx 1$–1.5 means "slow decay over 3–6 bars," not scale-free memory.

A.3 The one-lag baseline

Two short-memory rules are the natural sceptical benchmarks for a twelve-lag kernel, and we report both rather than leave them implicit: $N = 1$, betting against the previous candle, and $N = 3$, the shortest window at which the ranking gain is already complete. Rerunning the primary walk-forward with the constrained logit truncated to $N \in \{1, 2, 3, 6, 12\}$, on an identical universe, geometry, spin transform and selection protocol with only the kernel length changing, gives Table 5. At $N = 1$ the model is $\mathrm{sigmoid}(C + A s_{t-1})$, the probabilistic form of betting against the previous candle's sign (the fitted A is negative in 99% of pairs), and it already delivers a crypto class mean of 0.5267 (above 0.5 in 96% of pairs) against the twelve-lag 0.5305. Because both lag orders are scored on the same candles the increment is a paired quantity: bootstrapped jointly on the shared time grid, the crypto class-mean gain from $N = 1$ to $N = 12$ is +0.0039 (95% CI [+0.0026, +0.0052], $p < 10^{-3}$, positive in 92% of pairs): systematic but small, and essentially complete by $N = 3$ (+0.0039 already), consistent with the ~3-bar sign autocorrelation of Figure 4a. Log-loss and Brier move in the same direction and are likewise saturated by $N = 3$.

Nothing in Sections 3–5 turns on the kernel length: the class contrast is reproduced by the free logit, by R, by DFA, and by the $N = 1$ special case alike, with stock class means inside $[0.4978, 0.4993]$ at every N. The gap is a crypto-side phenomenon at every kernel length.

Why twelve, when three ranks as well? N is a measurement window, not a capacity parameter, and was fixed ex ante rather than read off this table; the kernel tie makes it nearly free (three parameters at every N), and sweeping $N \in \{6, \ldots, 48\}$ moves crypto-15 m AUC by less than 0.002 while the free AR(N) logit degrades monotonically. What the longer window carries is the decay estimate: the share of folds selecting α at an edge of its search grid falls from 64% at $N = 2$ to 30% at $N = 12$ (79–56% on the stock side, where there is no kernel to find). Ranking saturates by three lags because it ignores small tail weights; the tail is where the memory is measured.

A.4 Estimation and baselines

All models are fit by maximum likelihood: α profiled over a grid with (C, A) by unpenalized logistic MLE, selected by log-loss on a chronological validation tail (last 20% of each training window), then refit on the full window. The free,

	crypto (183)			stocks (187)			paired gain (crypto)		α (crypto)	
N	AUC	log-loss	Brier	AUC	log-loss	Brier	ΔAUC	% pairs > 0	median	% edge
1	0.5267	0.68238	0.24475	0.4978	0.68106	0.24399	—	—	—	—
2	0.5291	0.68220	0.24466	0.4983	0.68106	0.24399	+0.0025	85	1.24	64
3	0.5305	0.68209	0.24461	0.4982	0.68107	0.24399	+0.0039	95	1.16	51
6	0.5303	0.68212	0.24462	0.4987	0.68109	0.24400	+0.0037	91	1.23	37
12	**0.5305**	**0.68211**	**0.24462**	0.4993	0.68108	0.24400	**+0.0039**	**92**	1.27	**30**

Table 5: The constrained logit truncated to N lags, primary universe and protocol throughout. Columns 8–9: paired crypto class-mean AUC gain over $N = 1$ (joint-bootstrap CI at $N = 12$: $[+0.0026, +0.0052]$) and share of pairs positive. Last two columns: median fitted α and share of folds selecting α at an edge of its $[0, 3]$ grid (α not free at $N = 1$). Betting against the last candle captures most of the detection; the kernel adds a small increment saturating by $N \approx 3$.

ridge, and lasso AR(12) logits and the tuned machine-learning baselines select hyperparameters by the identical protocol, so competitors differ only in hypothesis space.

A.5 Why the constraint: variance reduction in a weak-signal regime

On a deeper within-crypto dataset (BTC/ETH/SOL/XRP, 5 m–4 h, BTC from 2024-03), the three-parameter model beats the unconstrained AR(12) logit on out-of-sample log-loss in 15/15 cells, with the largest gains in data-scarce cells: the free logit's train–test gap widens as the AR order grows while the constrained model's stays near zero. L_1/L_2 shrinkage barely moves the free model: norm penalties do not help a diffuse, ordered signal, whereas the *shape* prior does. Across folds the free AR(24) weight vector flips sign at a given lag in 30% of folds; the constrained kernel is sign-stable in 100%. Tuned gradient-boosted trees and an MLP on the same twelve lagged returns lose to the three-parameter model on out-of-sample log-loss in all cells, mirroring the shallow-beats-deep pattern in low-signal financial prediction [23, 33, 52]. Read these against the base rate, not against each other: an intercept-only predictor scores $\log 2 = 0.6931$, so on btc-1h the constrained model (0.6918) is the only one of the four with positive out-of-sample skill — the free logit (0.6935), GBM (0.6933) and MLP (0.7005) all score worse than predicting the base rate. This is why the free logit fails log-loss significance even on crypto-15 m while the constrained logit clears it: the two agree on AUC, which makes the contrast estimator-robust, and disagree on calibration, which is what the constraint buys.

Figure 9 shows the calibration claim directly rather than through aggregate score differences. On the pooled crypto out-of-sample predictions of the primary test (all 183 pairs, 5.1 million bar-level probabilities), the constrained model is calibrated essentially perfectly across its own prediction deciles, at a linear calibration slope of 1.00 (95% joint block-bootstrap CI $[0.95, 1.04]$), while the free logit over-disperses, with realized frequencies flatter than predicted at both tails (slope 0.82, CI $[0.75, 0.88]$, excluding 1). The pooled log-loss difference, free minus constrained, is +0.0015 nats (CI $[+0.0009, +0.0024]$), Brier concordant. Small in absolute units, but the yardstick is wrong: at AUC 0.531 the binormal discriminability is $d' = \sqrt{2}\,\Phi^{-1}(0.531) = 0.110$, and the entire log-loss reduction available to a perfectly calibrated forecaster is $d'^2/8 = 0.0015$ nats. The free logit's calibration deficit equals the total extractable signal — its extra capacity is spent manufacturing overconfident tails, the failure mode the three-parameter tie removes.

B Protocol, inference, and negative controls

B.1 Inference details

The primary test is boxed in Section 2; it was fixed in design before the wide universe was run (not a registered pre-analysis plan). Other horizons reuse the same geometry scaled in bars (1 h: 2,160/360; 4 h: 720/120). The joint bootstrap that produces the headline interval uses $B = 1{,}000$ resamples (`dependence`); raising it to $B = 5{,}000$ gives $[+0.028, +0.034]$ against the reported $[+0.027, +0.035]$, and blocks of 192/384/768 fifteen-minute slots at $B = 5{,}000$ give identical conclusions (`dependence_sensitivity`, `nonflat_gap`). The block bootstrap conditions on the fitted models, and two refit diagnostics quantify what that costs. A per-asset refit-inclusive bootstrap (50 assets, 25 per class, both models refit inside every resample, $B = 200$) widens intervals by a median factor of 1.12 and biases the classes in *opposite* directions: crypto −0.0069, stocks +0.0011, so the subset's class gap falls from +0.0278 to +0.0198 (29%). Because independent per-asset draws destroy the cross-sectional dependence, we also ran the joint version (`refit_joint`): one shared block draw per replicate, both models refit on all 28 assets (chosen at even AUC quantiles per class; the subset reproduces the universe, gap +0.0310 against +0.0313). It gives +0.0166 $[+0.0091, +0.0241]$, 54% retention. The two diagnostics bracket the correction — the fit-conditional headline is optimistic by a third to a half, putting the refit-corrected gap near +0.017 to +0.022. We do not restate the headline on subsets, and this correction is *not* independent of the artifact adjustments of Section 3, so the two must not be multiplied. The refit-inclusive joint SD (0.0038, against the fit-conditional 0.0018)

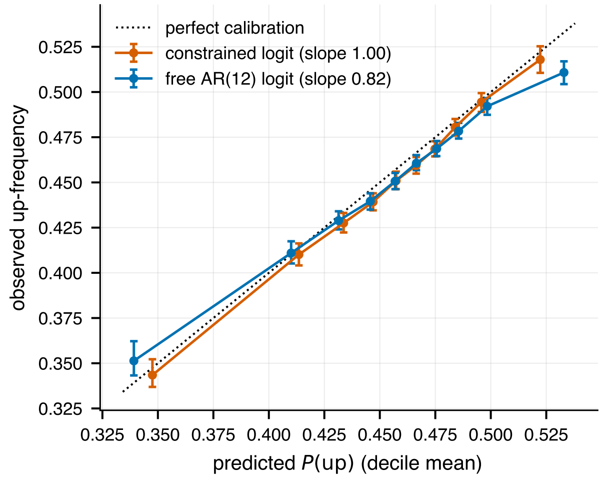

Figure 9: Reliability diagram, pooled crypto out-of-sample predictions (183 pairs): realized up-frequency by decile of predicted $P(\text{up})$, error bars from the joint block bootstrap. The constrained logit tracks the diagonal (slope 1.00, CI $[0.95, 1.04]$); the free AR(12) logit over-disperses (slope 0.82, CI $[0.75, 0.88]$).

doubles the interval width, the same message the permutation null below delivers from another direction.

Block *length* is common to the classes; block *count* is not. At 384 slots the median crypto pair supplies ~86 blocks against ~16 for the median stock, and a moving-block bootstrap on 16 blocks has variance-estimator relative error of order $\sqrt{2/16} \approx 35\%$ and under-covering percentile intervals — the per-stock half-widths (0.0185 vs. 0.0090 crypto) show it. Rerunning the per-asset test with block lengths scaled at the standard $n^{1/3}$ rate (anchored so the median crypto pair keeps 384 slots; the median stock then resamples ~28 blocks of 217 slots) leaves the discovery counts essentially unchanged: 165 of 183 crypto pairs and the same 5 of 187 stocks after joint BH (`blocklen_sensitivity`). The thin-block problem is therefore one of interval width, not of the tallies; it still applies untreated to the holdout stock leg (~5 blocks), whose interval should be read accordingly. The conjunction $p_\cap$ is a valid intersection–union p-value [6, 7], conservative under its null, so the BH step retains its guarantee.

Our "p" notation is also doing less than it appears to. The class-gap $p < 10^{-3}$ is the share of observed-centred bootstrap replicates at or below zero — a percentile-interval statement bounded below by $1/B$, and read best as "the 99.8% interval excludes zero"; the same applies to the DFA gap and the N=1 $\to$ N=12 increment. Per-asset p-values use $B = 300$ and cannot fall below 0.0033, a floor that sits near the BH threshold at the first two dozen ranks, so discovery counts are quantized exactly where marginal discoveries live.

B.2 An exact permutation null for the class gap

Since neither quantity above is a tail probability under a null, we construct one, and it is the inference we put first. A single circular time shift is applied to the label series of every asset simultaneously on the shared grid, destroying score–label alignment while preserving within-asset label autocorrelation and the entire cross-sectional dependence structure. Shifts are whole-day multiples (96 slots): listed instruments occupy 68 of the 96 slots-of-day against crypto's 96, and a day-aligned shift maps each asset's coverage onto itself, so no replicate changes class composition. All 340 admissible shifts are used.

The observed gap lies outside the entire null. Against a null centred at −0.0009 with SD 0.0073 and maximum +0.0165, the observed +0.0313 sits 4.4 null SDs out, reached by 0 of 340 shifts ($p = 0.0029$, the exact floor); the free logit reads 4.1 SDs, same p (`permutation_null`; an unrestricted-shift version agrees to three decimals). This null assumes no stationarity, symmetry or block length, and because it preserves the dependence exactly, the population claim no longer leans on BH-under-dependence. One diagnostic cuts the other way: the null's dispersion is four times the joint bootstrap's (0.0073 vs. 0.0018). The estimands differ, but the direction matches the refit widening above, and the reported $[+0.027, +0.035]$ is likely optimistic in width even though the gap's significance is not in doubt.

B.3 Power of the equity null

The power statement of Section 3 is computed entirely from the frozen per-asset out-of-sample series, with no new modeling. For each of the 187 stocks the moving-block bootstrap (block 384 slots, as in the per-asset test) gives the sampling dispersion $\hat{\sigma}_i$ of its out-of-sample AUC; under a normal approximation of the bootstrap law, the power of the one-sided 5% test against a true excess Δ is $\Phi(\Delta/\hat{\sigma}_i - z_{0.95})$. At $\Delta = 0.031$ (the crypto class mean) the per-stock powers are as quoted in Section 3, with power $\geq$ 0.9 in 78% of stocks (the weakest series has $n_{\mathrm{OOS}} \approx$ 3,000); the per-asset minimum excess detectable with 99% power is 0.033 at the median stock, which is why the class-level bound, whose power comes from pooling, carries the claim. The joint moving-block bootstrap of the stock class-mean AUC on the shared time grid (as in the primary gap inference) gives an observed mean of 0.499 with the one-sided 95% upper bound of 0.5007 quoted there, excluding any common equity-class effect larger than ≈0.0007 of AUC (`power`).

B.4 Negative controls

Shuffled labels destroy all feature–label dependence: on the focal panel, mean AUC 0.500 and 0/14 significant; at full scale (all 370 series through the identical pipeline, FDR step included), mean AUC 0.499 (crypto) and 0.496 (stocks), class gap +0.002 against the observed +0.031, and 0/370 significant after FDR. The machinery cannot manufacture the result. *Phase-randomized surrogates* (IAAFT) [47, 49] preserve linear autocovariance while destroying nonlinear dependence. On the focal coins the signal collapses (BTC 0.533 → 0.506), pointing to a predominantly nonlinear signal — BTC's $\rho_1 = -0.003$ supports only 0.501 through the linear channel of §3 against 0.533 observed. But the control itself is flawed, in three ways we report rather than smooth over. It is anti-conservative on session-heteroskedastic series: surrogates clear the two-model criterion on 7 of 14 instruments (5 of 11 non-crypto against 3 of 11 real), because IAAFT preserves ρ_1 while destroying the volatility clustering that makes it unrankable (SPX: $\rho_1 = -0.030$, real AUC 0.492, variance-homogenized 0.515); the free logit reaches a mean AUC of 0.520 on surrogates, bounding how much of the conjunction criterion's second leg is linear read-out. And at class level the arithmetic does not reconcile: the crypto class-mean $\rho_1 = -0.046$ implies a linear-channel AUC of 0.521 — two thirds of the excess — yet surrogates retain a fifth. The per-asset accounting resolves that discrepancy as compositional: across the 183 pairs, each asset's linear-implied excess $\Phi(\sqrt{4/\pi}\,|\rho_1|) - 0.5$ and its observed excess are essentially uncorrelated ($r = +0.06$; median crypto ρ_1 is -0.028 against the mean's -0.046), so the class-mean ρ_1 is carried by bounce-prone pairs whose AUC excess is no larger, and a class-mean linear implication does not describe any particular pair (`linear_channel`). The mirror image holds for stocks: their linear channel implies +0.015 of excess that never ranks (observed −0.001), the heteroskedasticity reading in cross-section. The IAAFT row bounds the linear channel and certifies nothing.

The null with correct size is the sign-randomization surrogate: hold each series' $|r|$ path fixed — volatility clustering and the diurnal profile survive *exactly* — and randomize only the signs, iid at the empirical up-rate, destroying precisely the dependence under test. Run on the fourteen focal instruments under the identical walk-forward and bootstrap, it is decisively clean: **0 of 14** clear the two-model criterion (crypto mean AUC 0.496, non-crypto 0.498), against IAAFT's 7 of 14 (`signrand`). The machinery finds nothing once sign dependence is removed, with every magnitude structure intact.

Time reversal, a third control, preserves the autocovariance exactly while destroying time-*asymmetric* structure, and it cuts against one of our readings. The crypto mean falls only 0.536 to 0.521 with all three coins still two-model significant and couplings still negative; the non-crypto side reads 0.501, 0 of 11. So the class contrast is robust to reversal — one more estimator-independent confirmation — but most of the detectable structure is time-symmetric, and the directional overreaction-then-correction story of Section 5 therefore rests on the flow conditioning, not on the sign path.

A generative check: a Glauber chain simulated from the fitted parameters reproduces the predictive level (AUC 0.535 vs. 0.538) but over-produces linear signatures (ρ_1 to −0.07 against an observed ≈ − 0.003, a >10-SE rejection at $n \approx 33{,}000$; VR(16) ≈ 0.84 vs. 0.97). It is an adequate predictive reduction and a decisively incomplete generative model — unsurprising, since Eq. (1) consumes a magnitude and emits a sign probability, so any simulation must supply a sign-to-return map the estimator never specified.

C Focal panel and horizon dependence

Under the two-model criterion the focal tally is crypto 3/3, FX 2/3, US-listed 1/8 (Table 6); permuting class labels over per-instrument AUCs, the crypto-minus-rest gap of +0.030 has $p = 0.003$. AAPL and NVDA carry the familiar small negative bid–ask-bounce coupling of individual equities [36, 39, 46] yet barely clear AUC 0.5; crypto's coupling is two-to-three times larger and strong enough to rank direction out of sample. Focal mean AUC by horizon (Figure 2b): crypto 0.536/0.521/0.496 at 15 m/1 h/4 h, FX 0.513/0.507/0.487, US-listed ≈0.50 throughout. By the bootstrap, crypto is significant in 4/4 coins at 1 h under the constrained logit with the free logit corroborating 3/4 (SOL from the independent-venue refetch, as at 15 m); FX drops to 0/3 at 1 h; US-listed is 0/8 at both 1 h and 4 h. At 4 h $n_{\mathrm{OOS}} \approx 10^3$ and the apparent crypto-reversal/ equity-momentum flip is suggestive only. Crypto coupling stays negative at 1 h ($\bar{A} \approx -0.13$) before fading toward zero at 4 h; US-listed drifts from weak reversal to weak momentum.

Instrument	Class	n_{OOS}	AUC	95% CI	Constr. p	Free p	A	ρ_1
ETH	crypto	33,312	0.538	[0.532, 0.544]	$\mathbf{< 10^{-3}}$	$\mathbf{< 10^{-3}}$	−0.31	+0.005
XRP	crypto	33,312	0.536	[0.530, 0.542]	$\mathbf{< 10^{-3}}$	$\mathbf{< 10^{-3}}$	−0.27	−0.023
BTC	crypto	33,312	0.533	[0.527, 0.539]	$\mathbf{< 10^{-3}}$	$\mathbf{< 10^{-3}}$	−0.33	−0.003
GLD	commodity	11,642	0.520	[0.506, 0.531]	**0.005**	$\mathbf{< 10^{-3}}$	−0.10	+0.024
GBPUSD	fx	21,928	0.516	[0.510, 0.523]	$\mathbf{< 10^{-3}}$	**0.002**	−0.74	−0.023
EURUSD	fx	21,930	0.514	[0.507, 0.523]	$\mathbf{< 10^{-3}}$	**0.014**	−0.71	−0.021
NVDA	stock	11,996	0.512	[0.501, 0.524]	0.015	0.258	−0.13	+0.013
USDJPY	fx	21,934	0.509	[0.502, 0.516]	0.008	0.239	−0.13	−0.010
AAPL	stock	11,940	0.505	[0.495, 0.514]	0.176	0.055	−0.13	−0.008
QQQ	index	11,994	0.501	[0.493, 0.510]	0.353	0.116	−0.07	−0.006
TSLA	stock	11,996	0.501	[0.490, 0.508]	0.506	0.211	−0.05	+0.007
IWM	index	11,975	0.497	[0.485, 0.507]	0.744	0.138	−0.05	−0.033
TLT	bond	11,917	0.492	[0.483, 0.502]	0.927	0.643	−0.07	+0.008
SPX	index	12,006	0.492	[0.482, 0.500]	0.972	0.641	−0.05	−0.030

Table 6: The focal panel at 15 m: out-of-sample AUC (constrained logit), 95% CI, one-sided p for AUC> 0.5 under both models, fitted coupling A (fixed λ; standardized values in Appendix D), and lag-1 return autocorrelation. Bold: $p < 0.05$ under both models. SOL lacks matched 15-minute data here; the independent-venue refetch fills the gap (AUC 0.529, both models $p < 10^{-3}$).

D Robustness battery

Hyperparameters and standardized spins Sweeping $\lambda \in \{50, \ldots, 1000\}$ and $N \in \{6, \ldots, 48\}$ moves mean crypto-15 m AUC by less than 0.002. The volatility-standardized spin leaves the picture unchanged (crypto 0.535, $\bar{A} = -0.32$; non-crypto 0.505, $\bar{A} = -0.07$) and shrinks the visually inflated fixed-λ FX couplings of Table 6 to −0.14; coupling magnitudes should be compared on the standardized scale.

Regular trading hours Restricting the eight US-listed focal instruments to 09:30–16:00 New York yields mean AUC 0.500 with 0/8 significant (gold's marginal 24-h significance disappears); the full 187-stock universe on RTH bars gives mean 0.498 with 1.1% significant. Crypto, having no session, is unchanged: on like-for-like active-session bars the contrast widens slightly.

Multi-year stability Extending the four focal coins to 2021: all 24 coin-years significant under both models (AUC 0.523–0.550), couplings negative in every cell, with a mild downward BTC drift consistent with adaptive-markets readings [31, 34, 37] (Figure 3 in the main text). With nothing fitted, the sign correlation R per calendar quarter is negative in all 21 crypto quarters and 83/84 coin-quarters, while the US-listed mean never leaves $[-0.020, +0.007]$. The equity counterpart (40 instrument-years, including 2022) yields mean AUC 0.501 with ≈chance-level isolated significances. Crypto AUC is also flat across trailing volatility terciles (0.536/0.539/0.534) and UTC sessions (0.531–0.542): a standing feature of the tape, not an episodic one.

Detrended fluctuation analysis As an estimator outside the conditional-model family, DFA-1 [30, 43] on all 370 return series (twelve log-spaced scales, 8–512 candles) gives crypto a class-mean scaling exponent of $H = 0.469$ (below 1/2 in 88.5% of pairs) against 0.510 for US-listed instruments (28.3%), the model-free counterpart of a negative coupling [28]; under the joint bootstrap (with H re-estimated on block-capped scales inside each replicate) the class gap is −0.049 $[-0.084, -0.012]$ (`dfa`, Figure 10). Three caveats keep this row modest. It is not an independent probe: negative short-lag autocorrelation mechanically depresses DFA-1 H, so antipersistence and a negative coupling summarize one autocovariance structure. The estimate is scale-range sensitive (full-sample gap −0.041 against block-capped −0.049, a swing comparable to the interval's distance from zero), and these near-martingale series need carry no scaling regime at all, so we attach no meaning to the level. And one day is 96 candles, inside the fitting range: session periodicity puts a crossover in the stock fluctuation functions that 24/7 crypto lacks, biasing a single-slope fit asymmetrically, in the direction that widens the gap. Rerunning the stock leg on session-restricted bars (09:30–16:00 weekdays, the overnight cycle spliced out) confirms the bias and sizes it: the stock class mean falls from 0.510 to 0.495, with the below-1/2 share rising from 28% to 61% (`dfa_rth`). So roughly a third of the 24-hour class gap was session periodicity, the like-for-like gap is nearer −0.026 than −0.041, and stocks are themselves weakly antipersistent intraday — consistent with their small negative bounce ρ_1. The ordering survives (crypto 0.469 remains below the session-restricted stock mean), and the row corroborates the ordering established by R, no more.

Microstructure artifacts The one-bar-gap test (predicting bar t from $t-2, \ldots, t-13$) retains BTC 0.514, ETH 0.519, XRP 0.513 (all $p < 10^{-3}$ both models); at population scale crypto moves 0.531→0.522 with 81% of pairs still significant, stocks 0.499 either way (Figure 11a). Excluding flat bars (median 16% of bars across wide US-listed instruments, 6% wide crypto, < 1% focal coins) moves wide crypto 0.531→0.520 and stocks 0.4993→0.4985, with 97% of pairs above

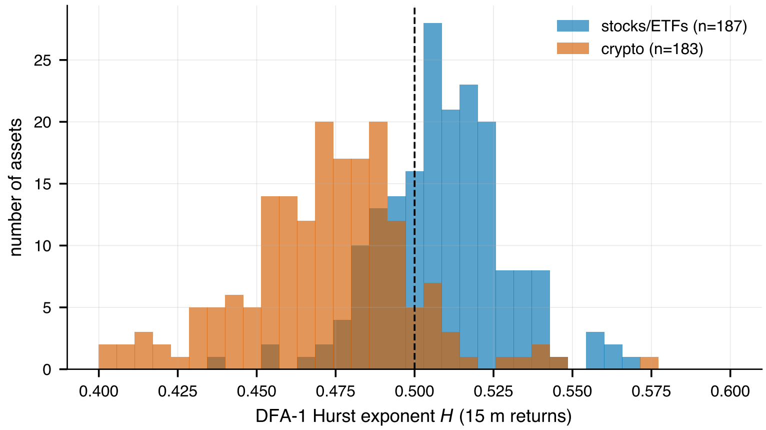

Figure 10: DFA-1 Hurst exponents at population scale, one count per asset over all 370: the crypto mass sits left of the random-walk value 1/2, the US-listed mass on it.

0.5. This is the preferred artifact-robust gap of Section 3, whose interval is computed under the identical joint bootstrap at $B = 5{,}000$ ($p < 2 \times 10^{-4}$; `nonflat_gap`). Accuracy metrics are far more sensitive to the flat-bar convention, which is why raw accuracy is never used as a headline metric. Stratifying by volume, the modest raw AUC–liquidity gradient is carried entirely by the flat-bar channel (non-flat quintile means 0.518–0.524, flat; Figure 11b); regressing flat-bar-robust AUC on log volume, flat share, and median $|r_t|$ (HC1), the intercept is 0.520 and the volume slope slightly positive, the opposite of a staleness artifact. (The raw ex-ante volume–AUC rank correlation of −0.16 and this conditional slope differ in sign; both are economically zero, and the ledger's "volume slope ≈0" refers to this conditional specification.)

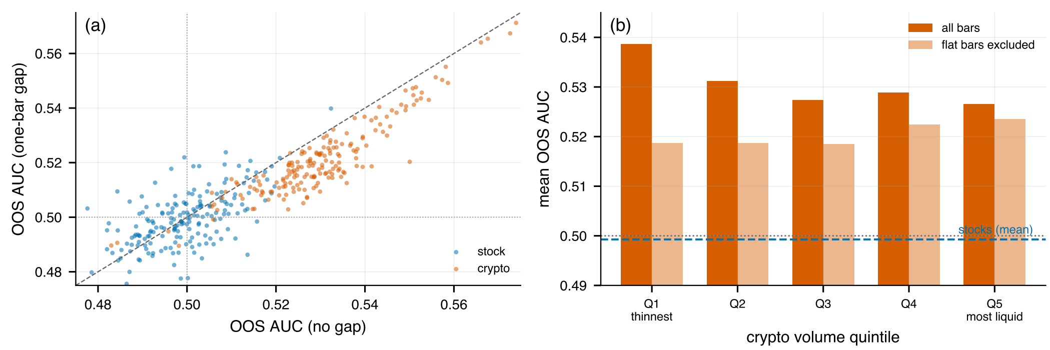

Figure 11: Microstructure-artifact tests at population scale. (a) Per-asset AUC with vs. without a one-bar gap between features and label. (b) Crypto AUC by volume quintile, all bars vs. non-flat bars: the thin-pair advantage is a flat-bar artifact; the artifact-free signal is uniform. Dashed line: stock mean; dotted line: 0.500.

Bar-grid alignment Quarter-hour clock effects, meaning order-flow or quoting seasonality concentrated at :00/:15/:30/:45, could in principle manufacture structure specific to the conventional 15-minute grid. Rebuilding bars from public 1-second klines with openings shifted to +1, +2, +5, and +7 minutes past the quarter-hour, the sign correlation R of Eq. (2) stays strongly negative at every offset on the three focal coins with 1-second archives (SOL lacks one): BTC −0.063 on-grid against −0.047 to −0.061 off-grid; ETH −0.094 against −0.058 to −0.073; XRP −0.075 against −0.050 to −0.073; all fifteen 95% block-bootstrap intervals exclude zero, and the on-grid row is rebuilt from that same archive under the same bar-completeness rules as the off-grid rows (2026-02–2026-05, ~9,400 bars per offset; ~4,400 for XRP, whose archive ends 2026-03). The signal does not depend on the conventional grid; the on-grid values are the largest in magnitude (most visibly for ETH), consistent with a small clock-aligned component on top of it. The check covers the focal coins, the instruments with public 1-second archives.

External validity Refetching the four focal coins from Coinbase, OKX, and Bybit and re-running everything: all 12 venue–coin cells significant under both models with per-coin AUC within ±0.002 of Binance. Dukascopy bid/ask-averaged quote-midprice bars, on which bid–ask bounce cannot exist, reproduce the effect (BTC 0.529, ETH 0.537, couplings ≈ − 0.30). Those bars carry a median quoted spread of 5.4 bp of mid for BTC and 17 bp for ETH against a thresholded gross edge of ~1 bp, but Dukascopy quotes a retail contract-for-difference, not the Binance book, where top-of-book on these pairs is of order a tick. We therefore use them for the bounce control rather than as an independent price of the friction, which is set by fees (Appendix G). Close-to-close labels coincide with open-to-close to the third decimal; a VWAP-to-VWAP variant scores higher only through the mechanical smoothing dependence of time-averaged prices [53] and is not tradable structure.

Universe selection The 183-pair universe was frozen on a post-sample volume ranking (survivor set; Appendix G). The ex-ante specification of Section 3, whose headline statistics are reported there, is recomputed by aggregation over the frozen per-asset outputs with no model refits (`exante_primary`); the counts behind the percentages are 129/133 vs. 5/187, the within-class mean pairwise correlation is 0.54 ($N_{\text{eff}} \approx 1.9$), and the ex-ante top-100 subset gives a gap of $+0.033$ $[+0.029, +0.037]$. Across ex-ante volume quintiles crypto mean AUC is flat (0.530–0.538).

E Wrapper panel: instrument-level detail

Underlying	Instrument	Role	AUC	non-flat	95% CI	conj. p	A
Gold	XAUUSD	underlying	0.503	0.503	[0.489, 0.518]	0.282	−0.28
	GLD	wrapper	0.507	0.506	[0.491, 0.520]	0.233	−0.20
	GDX	correlated	0.499	0.500	[0.483, 0.512]	0.605	−0.01
	NEM	correlated	0.501	0.500	[0.484, 0.515]	0.704	−0.02
Silver	XAGUSD	underlying	0.500	0.500	[0.482, 0.515]	0.546	−0.03
	SLV	wrapper	0.505	0.500	[0.488, 0.521]	0.276	−0.03
Platinum	XPTUSD	underlying	**0.532**	0.531	[0.517, 0.548]	**0.001**	−0.30
	PPLT	wrapper (thin)	0.511	0.505	[0.492, 0.528]	0.129	−0.06
Palladium	XPDUSD	underlying	**0.523**	0.521	[0.507, 0.536]	**0.006**	−0.21
	PALL	wrapper (thin)	0.510	0.502	[0.492, 0.526]	0.157	−0.10
Bitcoin	BTC	underlying	**0.524**	0.525	[0.513, 0.538]	**0.025**	−0.19
	IBIT	wrapper	**0.525**	0.524	[0.511, 0.540]	**0.004**	−0.20
	FBTC	wrapper	**0.525**	0.525	[0.513, 0.541]	**0.008**	−0.24
	BITB	wrapper	**0.525**	0.526	[0.512, 0.540]	**0.002**	−0.24
	ARKB	wrapper	**0.523**	0.523	[0.509, 0.537]	**0.009**	−0.22
	HODL	wrapper	**0.533**	0.532	[0.519, 0.546]	**0.000**	−0.25
	BTCO	wrapper	**0.516**	0.507	[0.502, 0.532]	**0.017**	−0.17
	EZBC	wrapper	**0.517**	0.514	[0.501, 0.532]	**0.026**	−0.22
	GBTC	wrapper (conv. trust)	**0.521**	0.520	[0.507, 0.536]	**0.005**	−0.19
	BITO	futures NAV	**0.524**	0.523	[0.512, 0.537]	**0.030**	−0.23
	BITU	2× lev. (deriv.)	**0.527**	0.527	[0.514, 0.542]	**0.004**	−0.15
	COIN	correlated	0.520	0.520	[0.506, 0.535]	0.364	−0.11
	MSTR	correlated	0.506	0.505	[0.490, 0.521]	0.248	−0.05
Ether	ETH	underlying	0.524	0.523	[0.507, 0.538]	0.150	−0.17
	ETHA	wrapper	0.522	0.522	[0.508, 0.539]	0.407	−0.16
	FETH	wrapper	0.522	0.522	[0.506, 0.536]	0.398	−0.17
	ETHE	wrapper (conv. trust)	0.525	0.524	[0.510, 0.539]	0.345	−0.16
	ETHU	2× lev. (deriv.)	0.522	0.522	[0.508, 0.537]	0.378	−0.12
Euro	EURUSD	underlying	0.504	0.504	[0.489, 0.518]	0.263	−0.57
	FXE	wrapper (thin)	0.484	0.493	[0.467, 0.501]	0.967	−0.25
	UUP	correlated	0.503	0.502	[0.489, 0.516]	0.472	−0.04
Yen	USDJPY	underlying	0.508	0.507	[0.494, 0.522]	0.148	+0.05
	FXY	wrapper (thin, inv.)	0.499	0.495	[0.486, 0.511]	0.638	+0.14

Table 7: Every panel leg on session-matched RTH bars (identical NY 09:30–16:00 slots, 2025-01-01 to 2026-02-11; identical walk-forward, models, bootstrap). "Non-flat" rescores with flat bars dropped; CI, p, A refer to the all-bars fit. Bold: two-model significant at raw 5% (within-panel FDR: §4.1). "Thin": $>$ 15% flat bars on the 24-hour tape (2.7–10.7% on these RTH slots). "Inv.": FXY moves inversely to the USDJPY quote; AUC and the sign of A are invariant under $r \to -r$.

Rescoring on non-flat bars moves ten of the seventeen wrappers by less than 0.001, and the exceptions are informative: the two thin metal wrappers fall further (PPLT 0.511 → 0.505, PALL 0.510 → 0.502), as do two of the eight Bitcoin funds (BTCO 0.516 → 0.507, EZBC 0.517 → 0.514), while FXE rises towards the band (0.484 → 0.493). The family readings of Table 3 survive it: the Bitcoin wrapper mean moves to 0.522 against an underlying of 0.525 (family gap −0.003 against −0.001) and the Ether mean to 0.523 against 0.523 (−0.001 against −0.000). On 24-hour bars, where their full sessions enter, all four spot metals are themselves two-model significant (AUC 0.513–0.522, $A = -0.18$ to -0.24): the OTC-metals analogue of the crypto finding, sitting between FX and crypto.

Table 7 gives every panel leg; the twelve event-study era cells are in Table 8 below.

Era	Span	n_{OOS}	flat	wrapper AUC	underl. AUC	gap	A_w / A_u
IBIT first 6 mo	2024-01 – 2024-07	1,690	1.1%	0.519	0.514	+0.005	−0.07 / −0.08
FBTC first 6 mo	2024-01 – 2024-07	1,690	1.0%	0.525	0.514	+0.010	−0.12 / −0.08
GBTC first 6 mo	2024-01 – 2024-07	1,690	1.3%	0.521	0.514	+0.007	−0.08 / −0.08
ETHA first 6 mo	2024-07 – 2025-01	1,742	2.3%	0.485	0.486	−0.001	−0.05 / −0.01
FETH first 6 mo	2024-07 – 2025-01	1,737	2.4%	0.483	0.486	−0.003	−0.06 / −0.01
ETHE first 6 mo	2024-07 – 2025-01	1,740	2.4%	0.472	0.486	−0.014	−0.04 / −0.01
BITO 2021–23	2021-10 – 2023-12	12,813	4.0%	**0.515**	0.514	+0.001	−0.15 / −0.14
BITO 2024–26	2024-01 – 2026-02	12,038	2.3%	0.507	0.508	−0.001	−0.11 / −0.11
IBIT full era	2024-01 – 2026-02	12,038	1.0%	0.508	0.508	+0.000	−0.11 / −0.11
FBTC full era	2024-01 – 2026-02	12,034	0.9%	0.507	0.508	−0.000	−0.12 / −0.11
GBTC full era	2024-01 – 2026-02	12,024	1.1%	0.509	0.508	+0.002	−0.12 / −0.11
ETHE full era	2024-07 – 2026-02	8,595	2.2%	0.511	0.508	+0.002	−0.12 / −0.12

Table 8: Wrapper event studies on session-matched RTH bars, each leg scored on its own bars (unpaired gaps; paired: Figure 6). n_{OOS} and "flat" describe the wrapper leg; A_w/A_u are fitted couplings. Bold: two-model significant at raw 5% (BITO 2021–23, conj. $p = 0.042$). Alpaca's feed begins at GBTC/ETHE's uplisting, so their trust eras are unobservable.

E.1 One-minute construction and diagnostics

Construction Both one-minute series carry bucket-start UTC timestamps (verified empirically: the bar labelled 09:30 New York has the opening minute's elevated variance), and bars are paired only on exactly equal epoch seconds, with no nearest-neighbor joining. Returns are intra-bar open-to-close, so consecutive bars do not overlap and the inter-bar gap is discarded; this attenuates rather than inflates displaced correlations, and it excludes the one place spot mechanically leads the ETF, the accumulated overnight move at the 09:30 open. Lagged pairs are formed only within runs of consecutive minutes, so no lag straddles a session boundary. The session filter excludes weekends, non-trading days (by intersection), and the post-13:00 minutes of the two NYSE early-close days in span; an earlier version admitted those 306 post-close minutes, and excluding them *raises* the contemporaneous correlation slightly. $n = 49{,}170$ aligned minutes in 127 daily runs.

Diagnostics Three checks on the headline numbers. *Alignment placebo:* deliberately shifting the IBIT series by ±1 minute before the join collapses the lag-0 correlation from 0.982 to ≈0 and relocates the peak to exactly $k = \pm 1$; the machinery resolves a one-minute offset, so the lag-0 peak is not an artifact of coarse joining. *Authenticity:* the two series are not accidental copies of one another, since the IBIT returns lie on a half-penny price grid whose implied price tracks IBIT's actual 2025–26 path, and the regression beta of IBIT on BTC is 1.006. *Staleness:* our archived one-minute records retain bar open and close only, so zero return is the staleness proxy; 4.3% of aligned IBIT minutes have exactly zero return, and excluding them moves the contemporaneous correlation to 0.983 and the largest displaced term to 0.013, changing nothing. Alpaca's bars also carry per-bar volume and trade count, which would give a sharper screen; given how little the zero-return exclusion moves the numbers we did not rerun on it.

Feed caveats The US leg is Alpaca's IEX feed, not consolidated SIP/NBBO data: IBIT's 09:30 bar opens at the first IEX print rather than the official auction open, the close misses the closing cross (the per-minute correlation degrades from 0.981 midday to ≈0.95 in the final minute), and minutes with no IEX print are absent rather than zero-filled, so the sample is IEX-active minutes. All of these degrade the measured contemporaneity, so the reported $r = 0.982$ is, if anything, a lower bound on the consolidated-tape value; a replication on SIP data is the natural upgrade and requires only a data purchase, not a design change.

F Mechanism details: subsumption and the flat channels

Subsumption The pooled feature-set comparison is Table 4 of the main text. Behind it, the lag-1 sign coefficient attenuates by roughly a quarter when imbalance enters and remains dominant. In the 183-pair cross-section, Δflip retains an independent loading on out-of-sample AUC (+0.004 per SD, $t = 3.0$, HC1) when dollar volume and median trade size enter jointly; trade size itself carries nothing ($t = 1.0$). HC1 treats the 183 pairs as independent draws, which at mean pairwise return correlation 0.40 they are not, so these t-statistics are optimistic in level; we read the sign and the ordering, not the significance.

Channels that are flat *Perp–spot arbitrage:* the reversal replicates fully on the perpetual-futures tape (AUC 0.531–0.538, $A = -0.21$ to -0.36) but is flat in the lagged |basis| (pooled tercile AUCs 0.533, 0.537, 0.536). If basis-closing flows generated it, it should strengthen when the basis is stretched. *Funding carry:* flat in the lagged funding rate (0.532, 0.537, 0.537); predictions in the hour containing a funding settlement are slightly *less* accurate (0.531 vs. 0.537).

Volatility: flat across trailing-volatility terciles. *Session:* the edge peaks in the most active hours (10:00–14:00 and 21:00 UTC) and sags through the Asian morning, so the signal lives on the active tape rather than in empty sessions.

G Data construction and universe details

Conventions All bars are stamped UTC with bucket-start timestamps on a shared grid; missing buckets are left absent, never filled. Labels are intra-bar open-to-close, so gaps, splits, and dividends never cross a bar boundary. Flat bars are retained (flat-bar-robust rescoring throughout; Appendix D).

Crypto (Binance) Spot klines from the public REST API; Binance emits a bucket for every grid slot, including ones with no trades, so the bar grid is complete and no bar is imputed. Flat bars are therefore real intervals, but on a universe of the highest-volume pairs they are almost entirely bars that traded and whose net move rounded to zero at the tick, not no-trade intervals, which is why the flat-bar channel is a labelling artifact rather than a liquidity measure. Taker imbalance uses the kline taker-buy base-volume field; perpetual legs from the USDT-M futures API; funding history from the funding-rate endpoint. The wide universe was selected on 2026-06-08 (post-sample) as the highest 24-hour quote-volume USDT spot pairs, excluding pegged-base pairs and leveraged tokens, requiring ≥5,000 bars in span: a survivor universe, with the stock list curated at the same date so the cross-class contrast is symmetric; Appendix D bounds the selection channels empirically.

Post-sample holdout accounting The holdout refetch of Section 3 covers 2026-02-12 to 2026-08-08 with relaxed minimum-bar cuts, disclosed in the output file: 4 of the 183 frozen pairs had been delisted by the refetch and appear only as absences, 6 of the remaining 179 fall short of the relaxed cut, leaving 173 crypto pairs scored on ∼11 folds each, while only 52 of the 187 stocks clear the cut. The model-free statistic needs no folds and covers all 179 surviving crypto pairs and all 187 stocks.

US-listed instruments (Alpaca) Fifteen-minute and one-minute bars (IEX feed), unadjusted, full 24-hour session as distributed; RTH cells restrict to 09:30–16:00 America/New_York, DST-aware. GBTC and ETHE traded OTC before their ETF conversions and the IEX feed carries no OTC prints, so both begin at uplisting (2024-01-11, 2024-07-23). The spot-Bitcoin funds all list from 2024-01-11 and the spot-Ether funds from 2024-07-23, so every one of them covers the 2025-01–2026-02 panel span in full; the launch-era cells of Table 8 score IBIT, FBTC and GBTC over their first six months of listed trading. BITU and ETHU obtain their 2× exposure through swaps and cash-settled futures rather than spot custody.

Spot metals and FX (Dukascopy) Bid-side 15-minute candles with closed-market bars (`high==low`, weekends/holidays) removed; platinum and palladium under the feed's commodity instrument ids. The same source supplies the bid- and ask-side series behind the quote-midprice robustness bars.

Transaction-cost bands The cost bands of Section 5.2 are round-trip spot figures: ≈5 bp at maker fees and 10–20 bp at taker fees, bracketing published retail and mid-tier schedules on the major venues over the sample. Fees rather than spread are what bind (§5.2); the wider Dukascopy quoted spreads of Appendix D are a retail contract-for-difference venue's and are not used to price this friction.

H Reproducibility

Every number, table and figure is produced by a pipeline of small Python modules, distributed as the public replication package linked in the data-availability statement: pipeline code, frozen symbol lists, fetch scripts rebuilding every input from public sources, frozen result files with SHA-256 hashes, and a README mapping each exhibit to a single `python` command. Bootstrap and permutation procedures are seeded; reruns are deterministic given identical inputs. Key modules: the wide-universe study (`wide`, `signlag`, `dfa`, `exante_primary`, `nbaseline`, `reliability`, `power`, `holdout`, `nonflat_gap`, `bargrid`, `permutation_null`, `factor_variance`, `holdout_did`, `joint_artifact`, `refit_joint`, `blocklen_sensitivity`, `linear_channel`, `signrand_control`, `dfa_rth`); the wrapper panel (`panel`, `family_inference`, `panel_fdr`, `wrapper_events`, `wrapper_gap_boot`, `leadlag`); mechanism (`flow_test`, `flow_boot`, `flow_cross`, `perp_test`); and economics (`selective`).